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1 Résumé

Dans ce rapport, nous nous intéressons au modèle du polymère dirigé en environnement aléatoire (DPRE en anglais). Ce modèle consiste à considérer une marche aléatoire simple sur le réseau $\mathbb{Z}^d$, dont la loi est donnée par

$$S_0 = 0 \quad \text{a.s.}, \quad P(S_{n+1} = y \mid S_n = x) = \frac{1}{2d} \mathbf{1}_{|x-y|_1=1}.$$

Ensuite, indépendamment, un poids aléatoire $\omega(n, x)$ est attribué à chaque point (n, x) de l'espace-temps (ou plutôt temps-espace, en accord avec la convention) $\mathbb{N} \times \mathbb{Z}^d$. La famille $(\omega(n, x))_{(n, x) \in \mathbb{N} \times \mathbb{Z}^d}$ des poids est une famille de variables aléatoires réelles i.i.d., et l'on souhaite alors quantifier une influence de ces poids sur notre marche aléatoire, qui agissent comme des obstacles ou des récompenses. On définit alors, pour tout $n \geq 1$, une nouvelle mesure de probabilité sur l'ensemble des chemins $(\mathbb{Z}^d)^{\mathbb{N}}$ par :

$$P_n^\beta(dS) \propto e^{\beta \sum_{i=1}^n \omega(i, S_i)} P(dS).$$

Dans cette formule, on constate que l'influence de l'environnement sur le polymère (la marche aléatoire de loi P_n^β) est quantifiée par l'exponentielle du coût total des n premiers pas, multipliée par un paramètre $\beta \geq 0$. On s'intéresse alors naturellement au comportement limite de la mesure P_n^β lorsque $n \rightarrow +\infty$. L'enjeu est d'établir si l'environnement influence suffisamment le polymère pour rendre son comportement radicalement différent de celui de la marche aléatoire simple.

Comme couramment pour un tel modèle de physique statistique, on observe l'existence d'une transition de phase pour le paramètre β , c'est-à-dire l'existence d'une valeur critique β_c telle que pour $\beta < \beta_c$, l'environnement n'influence pas suffisamment le polymère (son comportement est comparable à celui d'une marche aléatoire simple), et pour $\beta > \beta_c$, le parcours du polymère est davantage déterminé par l'environnement que par sa loi initiale en espace vide. Dès lors, beaucoup de questions se posent sur cette transition de phase, notamment sur le comportement du polymère à la valeur critique β_c . Ce problème a été traité très récemment par Stefan Junk and Hubert Lacoin.

Durant ce stage, nous nous sommes intéressés au modèle du polymère dirigé défini pour une marche aléatoire initiale qui n'est pas forcément la marche simple sur $\mathbb{Z}^d$, mais n'importe quelle marche aléatoire symétrique sur un graphe infini. Cette extension du modèle n'a été traitée que par très peu d'articles, et si la valeur de β_c dépend évidemment de la géométrie du graphe (comme c'est déjà le cas pour $\mathbb{Z}^d$ selon la dimension $d \geq 1$), l'influence de cette dernière sur le comportement du polymère en β_c reste un problème ouvert. Ainsi, notre travail a consisté à essayer de généraliser les résultats de Stefan Junk et Hubert Lacoin à d'autres graphes et marches aléatoires.

Si quelques résultats se généralisent aisément, la plupart se sont révélés extrêmement dépendants de la symétrie du problème, que ce soit l'invariance par translation du réseau $\mathbb{Z}^d$ ou les probabilités de transitions égales sur chaque arête. Étendre les résultats désirés s'est alors montré difficile dans le temps imparti du stage. Cependant, nous avons tout de même pu donner des résultats pour une classe de graphes assez restrictive mais tout de même intéressante. Nous donnons également une explication des difficultés rencontrées et dressons un état de l'art sur l'ensemble des questions relatives au problème, que ce soit pour le modèle défini sur $\mathbb{Z}^d$ ou sur un graphe quelconque.

2 Abstract

In this report, we study the Directed Polymer in a Random Environment (DPRE). This model is defined by considering a simple random walk on the lattice $\mathbb{Z}^d$, whose law is given by

$$S_0 = 0 \quad \text{a.s.}, \quad P(S_{n+1} = y \mid S_n = x) = \frac{1}{2d} \mathbf{1}_{|x-y|_1=1}.$$

Independently, a random weight $\omega(n, x)$ is assigned to each point (n, x) of space-time (or time-space, in accordance with the convention) $\mathbb{N} \times \mathbb{Z}^d$. The family of weights $(\omega(n, x))_{(n, x) \in \mathbb{N} \times \mathbb{Z}^d}$ consists of real-valued

i.i.d. random variables. We then seek to quantify the influence of these weights on the random walk, as they may act either as obstacles or as rewards. We define, for each $n \geq 1$, a new probability measure on the set of paths $(\mathbb{Z}^d)^\mathbb{N}$ by

$$P_n^\beta(dS) \propto e^{\beta \sum_{i=1}^n \omega(i, S_i)} P(dS).$$

In this formula, the influence of the environment on the polymer (the random walk whose law is P_n^β) is quantified through the exponential of the total energy collected during the first n steps, multiplied by a parameter $\beta \geq 0$. We are then naturally interested in the limiting behavior of the measure P_n^β as $n \rightarrow +\infty$. The main question is whether the influence of the environment is significant enough to make the behaviour of the polymer genuinely different from that of the simple random walk.

As is common for such models in statistical physics, one observes the existence of a phase transition in the parameter β , namely a critical value β_c such that, for $\beta < \beta_c$, the environment does not influence the polymer strongly enough, so that its behavior remains comparable to that of the simple random walk, whereas for $\beta > \beta_c$, the polymer path is determined more by the environment than by its original law in an empty space. This naturally raises many questions on this phase transition, in particular regarding the behavior of the polymer at the critical value β_c . This problem has been investigated very recently by Stefan Junk and Hubert Lacoin.

During this internship, we studied the directed polymer model when the underlying random walk is not necessarily the simple random walk on $\mathbb{Z}^d$, but rather an arbitrary symmetric random walk on an infinite graph. This extension of the model has been considered only in a small number of papers. While the value of β_c obviously depends on the geometry of the underlying graph, as is already the case for $\mathbb{Z}^d$ depending on the dimension $d \geq 1$, the influence of the geometry on the behavior of the polymer at β_c remains an open problem. Our work therefore aimed at generalizing the results of Stefan Junk and Hubert Lacoin to other graphs and random walks.

Although some of the results generalize relatively easily, most of them turned out to rely heavily on the symmetries of the problem, whether through the translation invariance of the lattice $\mathbb{Z}^d$ or through equal transition probabilities along each edge. Extending the desired results therefore proved difficult within the time available for the internship. Nevertheless, we were able to obtain results for a rather restrictive but still interesting class of graphs. We also provide an account of the difficulties encountered and give an overview of the current state of the art on the various questions surrounding the problem, both for the model defined on $\mathbb{Z}^d$ and for the model on a general graph.

3 Introduction and context

The Directed Polymer in a Random Environment (DPRE) is a statistical mechanics model introduced by the physicists David A. Huse and Christopher L. Henley in [HH85]. Their initial objective was to describe the interfaces of the planar Ising model with random coupling constants at low temperature. More generally, from a physical perspective, the DPRE model aims to understand the shape of polymer chains evolving in a medium filled with impurities. It falls into the broader category of disordered systems, where an initial system is perturbed by an external force. Here, the initial system is a simple random walk $(S_n)_{n \geq 0}$ in the lattice $\mathbb{Z}^d$, and the external force is a random environment, i.e. a family $(\omega(n, x))_{(n, x) \in \mathbb{N} \times \mathbb{Z}^d}$ of real-valued *i.i.d.* random variables so that $\omega(n, x)$ acts as an obstacle or a reward for the system. In order to model the influence of this medium on the polymer, the probability, for the simple random walk, of choosing a certain path is multiplied by

$$\frac{1}{Z_n^\beta} \exp(\beta H_n(\omega, S)) ,$$

where $H_n(\omega, S) := \sum_{i=1}^n \omega(i, S_i)$ is the energy of the n first steps of the path, called the Hamiltonian function, $\beta \geq 0$ is a parameter, interpreted as the inverse temperature, and Z_n^β is a normalizing constant. The new probability measure thus constructed is called the polymer measure. The choice of the map $E \mapsto \exp(\beta E)$ to quantify the influence of the environment as a function of energy is quite natural, since it makes the polymer measure a Gibbs measure. These measures are often considered in statistical physics as they maximize entropy at a given energy.

Afterwards, the model has gained a lot of interest from the mathematics community, partly because of its connection with the stochastic heat equation with multiplicative noise and the KPZ universality class. The starting point of the research is the existence of a critical inverse temperature, denoted β_c , which marks a phase transition: before it, the polymer is not sufficiently influenced by its environment and behaves similarly to the simple random walk, and after it, the environment becomes the principal factor determining the paths of the polymer. Many questions arise regarding this phase transition. The first one concerns the non-triviality of the phase transition ($\beta_c \neq 0$) and has been treated in [CH02, CSY03, CY06], or follows from stronger results given in [Bol89, Lac10]. A second objective is to characterize the difference in behaviour of the polymer between its two phases [CSY03]. Another question is the behaviour of the polymer at the critical inverse temperature β_c : this problem has been treated in many works under some conditions on the environment, but has been answered in all generality in some very recent works [Jun21, JL24, JL25, Jun25, JL26]. These three problems are among the most important, although many others have been studied. An extensive survey has been conducted in the lecture notes of Francis Comets from St. Flour's 2016 Summer Probability School [Com17].

As mentioned in the abstract, the objective of this internship has been to consider the polymer model for general random walks on infinite graphs. Some articles have been interested in the model outside of its usual frame, and have defined it on trees [BPP93], the discrete torus [EW89] and complete graphs [CMR19]. Only recently have two articles [KKN20, CSZ21] been interested in defining the model for general random walks and graphs, in order to understand which of their properties matter for the model. Our personal objective has been to continue the works of these articles by trying to extend the results of [JL25] to general infinite graphs.

This report is organized as follows. After giving our notation and some reminders on random walks on graphs in Sections 4 and 5, we will define the directed polymer model on general graphs and present the phase transition Theorem and the objects of interest for studying the model (Section 6). Then, in Section 7, we will summarize the state of the art on the problems mentioned earlier and state the known results, both for the usual setting of the simple random walk in $\mathbb{Z}^d$ and for the general setting. A particular emphasis on the results of [JL25, JL26] will be made in Section 7.3. Then, in Section 8, we present our developments in the effort of extending these results to general graphs, before concluding in Section 9. Finally, Appendix A gathers some classical results in probability theory, while Appendix B details the proofs of some of our statements.

4 Notation

- $\mathbb{N} = \{0, 1, \dots\}$ is the set of natural numbers, including zero.
- The power set of a set V is denoted by $\mathcal{P}(V)$, while for any $k \geq 1$, $\mathcal{P}_k(V) := \{X \in \mathcal{P}(V) \mid \#X = k\}$ is the set of all subsets of V whose cardinality is k .
- The expected value of a random variable X on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is denoted by $\mathbb{P}[X]$.
- Given a countable set D , $\mathcal{M}_1(D)$ denotes the set of probability measures on the measurable space $(D, \mathcal{P}(D))$.
- $\|\cdot\|_1$ denotes the taxicab norm on $\mathbb{R}^d$.
- Two functions $f, g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ are said to be comparable, written $f(u) \asymp g(u)$, if there exist $C > 1$ and $u_0 \geq 0$ such that:

$$\forall u \geq u_0 \quad \frac{g(u)}{C} \leq f(u) \leq Cg(u) .$$

- Given two random variables X and Y , we denote by $X \stackrel{d}{\sim} Y$ when they have the same distribution.
- The normal distribution of mean $\mu \in \mathbb{R}$ and standard deviation $\sigma \geq 0$ is denoted by $\mathcal{N}(\mu, \sigma)$. The Bernoulli distribution of parameter $p \in [0, 1]$ is denoted by $\mathcal{B}(p)$.

5 Some reminders on random walks on weighted graphs

In this section, we recall some basic definitions on random walks on graphs, so that the different cases later studied can be clearly stated.

Definition 5.1. A *(non-oriented) graph* is a collection $G = (V, E)$ of a set V , called the set of *vertices*, assumed to be countable in this report, and a subset $E \subseteq \mathcal{P}_2(V)$ called the set of *edges*. We write $x \sim y$ when $\{x, y\} \in E$.

A graph G is naturally equipped with the *graph distance*:

$$d(x, y) := \inf \{n \geq 0 \mid \exists x_0, \dots, x_n \in V \quad x_0 = x, x_n = y, x_i \sim x_{i+1}\}$$

The ball of radius $n \in \mathbb{N}$, centered at $x \in V$, is denoted by $B(x, n) := \{z \in V \mid d(x, z) \leq n\}$.

Definition 5.2. The direct *neighborhood* of a vertex $x \in V$ is the set:

$$N(x) := \{y \in V \mid y \sim x\} .$$

In this study, G is assumed to be *connected* and *locally finite*, which means that for any $x, y \in V$:

$$d(x, y) < +\infty \quad \text{and} \quad \#N(x) < +\infty . \quad (\text{CLF})$$

In the following sections, we will consider random walks on graphs. To simplify the statements, instead of considering an arbitrary random walk on a graph, the graph itself will be endowed with weights, so that one can consider a random walk with transition probabilities naturally given by these weights.

Definition 5.3. A *weighted graph* (G, μ) is a graph $G = (V, E)$, equipped with a family of *weights* $\mu = (\mu_{x,y})_{x,y \in V} \in \mathbb{R}^{V \times V}$, satisfying, for any $x, y \in V$:

$$\mu_{x,y} = \mu_{y,x} \quad , \quad \mu_{x,y} \geq 0 \quad , \quad \forall x \neq y : \mu_{x,y} > 0 \iff x \sim y .$$

Remark 5.4. Notice that we allow $\mu_{x,x} > 0$. Therefore, weights are put on all edges, and possibly on some vertices: for the other pairs of distinct vertices, the weights are null.

The weights can be interpreted as electrical conductance in electrical engineering terminology. We also define the total conductance of a vertex $x \in V$ as:

$$\mu_x := \sum_{y \in V} \mu_{x,y} ,$$

where the sum is actually finite, since the weights are null outside $N(x) \cup \{x\}$, which is a finite set under the hypotheses (CLF).

Given a weighted graph (G, μ) , a random walk $S = (S_n)_{n \geq 0}$ is naturally defined on G , considering the canonical Markov chain associated with the transition matrix:

$$P = (p_{x,y})_{x,y \in V} = \left(\frac{\mu_{x,y}}{\mu_x} \right)_{x,y \in V} .$$

Therefore, fixing a starting point $x \in V$, $(S_n)_{n \geq 0}$ is a random walk with associated probability P_x such that:

$$S_0 = x \quad P_x - \text{a.s.} \quad \text{and} \quad \forall y, z \in V \quad P_x(S_{n+1} = z \mid S_n = y) = \frac{\mu_{y,z}}{\mu_y} .$$

The Markov chain S is irreducible, since the graph is connected and the conductance of any edge is positive. S is also reversible with respect to the measure $(\mu_x)_{x \in V}$, since $p_{x,y} \mu_x = \mu_{x,y} = \mu_{y,x} = \mu_y p_{y,x}$. A natural choice for the conductance is to set $\mu_{x,y} = \mathbb{1}_{x \sim y}$ for any pair $(x, y) \in V^2$. In that case, S is the

Simple Random Walk (SRW) on G with transition probabilities $p_{y,z} = \frac{1}{\#N(y)}$. By an abuse of notation, we will sometimes refer to the weighted graph $(G, (\mathbb{1}_{x \sim y})_{x,y \in V})$ simply as G .

As mentioned in the introduction, the directed polymer model has been studied almost exclusively in the case where $(S_n)_{n \geq 0}$ is the SRW in $\mathbb{Z}^d$, which is naturally equipped with a graph structure where $x \sim y$ if, and only if $|x - y|_1 = 1$. In that case, the starting point can be assumed to be $x = 0$ without loss of generality, as $\mathbb{Z}^d$ is invariant under translation (and the conductance is constant on the graph). Then, $(S_n)_{n \geq 0}$ satisfies:

$$S_0 = 0 \quad P_0 - \text{a.s.} \quad \text{and} \quad \forall y, z \in \mathbb{Z}^d \quad P_0(S_{n+1} = z \mid S_n = y) = \frac{1}{2d} . \quad (5.1)$$

To finish this part, let us define, for any $x \in V$ and $n \in \mathbb{N}$:

$$R_n(x) := \{y \in V \mid P_x(S_n = y) > 0\}$$

the set of all reachable vertices, in n steps from x . Since the transition probabilities are positive along all edges, it holds that:

$$B(x, n) \subseteq R_n(x) \cup R_{n-1}(x) \subseteq B(x, n) \cup \{x\} . \quad (5.2)$$

6 Directed polymer on a graph, general framework

6.1 Definition of the model

Let us now define the *Directed Polymer in a Random Environment (DPRE)* model. The random walk we defined on (G, μ) will be modified to take into account the influence of the environment. All vertices of G will be endowed with a random energy (or potential), which can be positive or negative and represent either a reward or an obstacle for the random walk. These potentials will be random and evolve over time.

Definition 6.1. Let $\Omega := \mathbb{R}^{\mathbb{N} \times V}$ and $\mathcal{F} := \bigotimes_{\mathbb{N} \times V} \mathcal{B}(\mathbb{R})$. A *random environment* is a probability measure Q on $(\Omega, \mathcal{F})$ such that the coordinate process $\omega = \{\omega(n, x) \mid n \in \mathbb{N}, x \in V\}$ is a set of non-constant, i.i.d real-valued random variables with a well-defined Laplace transform:

$$\forall \beta \in \mathbb{R} \quad e^{\lambda(\beta)} := Q[e^{\beta \omega(n, x)}] < +\infty . \quad (6.1)$$

For a given point $(n, x) \in \mathbb{N} \times V$ in time-space, $\omega(n, x)$ is called the potential of the vertex x at time n . Now that random media are defined, the simple random walk can be modified to define the *polymer measure*.

Definition 6.2. Let $S = (S_k)_{k \geq 0}$ be the natural random walk on a weighted graph (G, μ) . Given a realization ω of the random environment, the *Hamiltonian* of S at time $n \in \mathbb{N}$ is the energy of the path $(S_k)_{1 \leq k \leq n}$:

$$H_n(\omega, S) := \sum_{i=1}^n \omega(i, S_i) .$$

Definition 6.3. Let (G, μ) be a weighted graph, and $x \in V$ be a starting point. Let S be the natural random walk on (G, μ) starting at x , ω a realization of the random environment, and $n \in \mathbb{N}$. The *polymer measure* is a probability measure $P_{n,x}^\beta$ defined on $V^\mathbb{N}$ as:

$$P_{n,x}^\beta(dS) := \frac{1}{Z_{n,x}^\beta} e^{\beta H_n(\omega, S)} P_x(dS) , \quad (6.2)$$

where $Z_{n,x}^\beta := P_x[e^{\beta H_n(\omega, S)}]$ is a normalizing constant that makes $P_{n,x}^\beta$ a probability measure, called the *partition function*.

Hence, the probability of choosing a specific path is modified to take into account the energetic cost of the first n steps. Paths maximizing $H_n(\omega, S)$ are preferred, whereas paths minimizing it tend to be avoided by the polymer. This preference for high-energy paths is quantified by a parameter $\beta \geq 0$ that can be interpreted as the inverse temperature. Notice that from a physical perspective, one would rather favor paths with the lowest energy, thus defining the polymer measure with H_n replaced by $-H_n$. However, this difference in physical perspective has no impact on the mathematical study of the model, therefore the convention chosen for (6.2) is the most standard.

Let us stress that the choice for the exponential function is arbitrary, and one could model the influence of the environment by setting $P_{n,x}^\beta(dS) \propto \beta H_n(\omega, S) P_x(dS)$ for instance. Nevertheless, the choice of an influence quantified as an exponential function of the energy is quite natural from a statistical mechanics perspective. Indeed, this choice makes $P_{n,x}^\beta$ a Gibbs measure so that it maximizes the entropy of the path at a given energy.

6.2 The phase transition and its related questions

Now, the natural question to ask is how different the paths chosen by the polymer are from those chosen by the initial random walk ? In other words, how much does the environment affect the random walk ? In order to formulate our expectations, let us consider the case where $(S_n)_{n \geq 0}$ is the SRW in $\mathbb{Z}^d$ as in (5.1):

- If the dimension is large and the inverse temperature $\beta \geq 0$ is small, the environment should not greatly affect the random walk. Indeed, many obstacles will be negligible and the random walk will have many options to avoid the most extreme obstacles.
- If the dimension is small or the inverse temperature is large, the behavior of the polymer will be drastically different from the one of the initial random walk. The polymer will choose a particular high energy path with high probability.

To prove (or disprove) these expectations, a fruitful approach has been to study the behavior $Z_{n,x}^\beta$ as $n \rightarrow +\infty$. First, notice that while $Z_{n,x}^\beta$ is a constant making $P_{n,x}^\beta$ a probability measure under a fixed realization of the environment, it remains a random variable with respect to the environment, that is on $(\Omega, \mathcal{F}, Q)$. Hence, let us define the *normalized partition function* as:

$$W_{n,x}^\beta := \frac{Z_{n,x}^\beta}{Q[Z_{n,x}^\beta]} = P_x \left[e^{\beta H_n(\omega, S) - n\lambda(\beta)} \right]. \quad (6.3)$$

Remark 6.4. From the definition of $\lambda(\beta)$ (6.1), the independence of the coordinates of the environment, and Fubini's theorem, it holds that for any $x \in V$, $n \in \mathbb{N}$ and $\beta \in \mathbb{R}$: $Q[Z_{n,x}^\beta] = e^{n\lambda(\beta)}$.

Then $P_{n,x}^\beta(dS) = \frac{1}{W_{n,x}^\beta} e^{\beta H_n(\omega, S) - n\lambda(\beta)} P_x(dS)$, and studying the limit of $W_{n,x}^\beta$ is feasible from the following proposition:

Proposition 6.5. $W_{n,x}^\beta$ is a positive martingale on $(\Omega, \mathcal{F}, Q)$ with respect to the filtration $(\mathcal{F}_n)_{n \geq 0}$ defined by:

$$\forall n \in \mathbb{N} \quad \mathcal{F}_n := \sigma \{ \omega(i, x) \mid 1 \leq i \leq n, x \in V \},$$

where in particular $\mathcal{F}_0 = \sigma \{ \emptyset \} = \{ \emptyset, \Omega \}$.

Thus, $(W_{n,x}^\beta)_{n \geq 0}$ converges a.s. to a nonnegative limit denoted by $W_{\infty,x}^\beta := \lim_{n \rightarrow \infty} W_{n,x}^\beta$.

Proof. Let us prove that $(W_{n,x}^\beta)_{n \geq 0}$ is a martingale for any $x \in V$ and $\beta \geq 0$. Since positivity is immediate, the last statement will follow. Let $n \in \mathbb{N}$. From its definition (6.3), $W_{n,x}^\beta$ is $\mathcal{F}_n$ -measurable and is integrable

$(Q[W_{n,x}^\beta] = 1)$. Moreover:

$$\begin{aligned}
Q[W_{n+1,x}^\beta \mid \mathcal{F}_n] &= e^{-(n+1)\lambda(\beta)} Q \left[P_x \left[\prod_{i=1}^{n+1} e^{\beta\omega(i,S_i)} \right] \mid \mathcal{F}_n \right] \\
&= e^{-(n+1)\lambda(\beta)} P_x \left[\prod_{i=1}^n e^{\beta\omega(i,S_i)} Q[e^{\beta\omega(n+1,S_{n+1})}] \right] \quad (\text{Fubini's theorem and } (\omega(k,y)) \text{ i.i.d.}) \\
&= e^{-n\lambda(\beta)} P_x \left[\prod_{i=1}^n e^{\beta\omega(i,S_i)} \right] = W_{n,x}^\beta,
\end{aligned}$$

so $(W_{n,x}^\beta)_{n \geq 0}$ is indeed a martingale. □

Remark 6.6. Note that $W_{0,x}^\beta = 1$ a.s.

Given a positive martingale (which is then converging a.s.), the natural question is whether the limit is null or not. In our case, we will prove that the event $\{W_{\infty,x}^\beta > 0\}$ is a tail event for the filtration $(\mathcal{F}_n)_{n \geq 0}$, therefore Kolmogorov's zero-one law yields $Q(W_{\infty,x}^\beta > 0) \in \{0, 1\}$. Let us state what these two possible values would represent for the model:

- If $Q(W_{\infty,x}^\beta > 0) = 1$, then for n large enough, $W_{n,x}^\beta$ stays bounded away from 0. Consequently, $P_{n,x}^\beta(dS) = C(S)P_x(dS)$ where $C(S) = C_{n,\beta,\omega}(S) = (W_{n,x}^\beta)^{-1}e^{\beta H_n(\omega,S) - n\lambda(\beta)}$ is positive a.s. Therefore, the polymer measure is asymptotically absolutely continuous with respect to the initial random walk measure. In that case, the polymer is merely influenced by its medium and behaves similarly to the initial random walk.
- Otherwise, if $Q(W_{\infty,x}^\beta > 0) = 0$, then $W_{\infty,x}^\beta = 0$ a.s. and the two measures become harder to compare. The polymer is significantly influenced by its environment.

This preliminary analysis motivates the following terminology:

Definition 6.7. The set $\{\beta \geq 0 \mid Q(W_{\infty,x}^\beta > 0) = 1\} =: \mathbf{WD}$ is called the *Weak Disorder (WD)* region.

The set $\{\beta \geq 0 \mid Q(W_{\infty,x}^\beta > 0) = 0\} =: \mathbf{SD}$ is called the *Strong Disorder (SD)* region.

Remark 6.8. $\mathbf{WD} \ni 0$ but $\mathbf{SD}$ can be the empty set. Also, while $\mathbf{WD}$ and $\mathbf{SD}$ depend *a priori* on the starting point $x \in V$, Theorem 6.9 will show that this is, in fact, not the case.

The weak and strong disorder regions (or regimes) correspond to the intuition we formulated for the expectations on the model. Hence, one should anticipate weak disorder to hold for small values of β , and eventually strong disorder to hold for large values of β . The following theorem confirms this intuition and exhibits what is perhaps the most interesting property of the model, namely the existence of a phase transition.

Theorem 6.9 ([CSZ21, Proposition 1.2]). *Let $\beta \geq 0$. Either weak disorder holds for every $x \in V$ ($W_{\infty,x}^\beta > 0$ a.s.), or strong disorder holds for every $x \in V$ ($W_{\infty,x}^\beta = 0$ a.s.). Furthermore, there exists a critical inverse temperature $\beta_c \in [0, +\infty]$ such that for any $x \in V$:*

$$Q(W_{\infty,x}^\beta > 0) = \begin{cases} 1 & \text{for } \beta < \beta_c. \\ 0 & \text{for } \beta > \beta_c. \end{cases}$$

This theorem was originally proved in [CY06] for the case of the SRW in $\mathbb{Z}^d$ and represents the starting point for the martingale approach on the model. It is easily generalized to any weighted graph that satisfies (CLF), and such a generalization has been done in [CSZ21]. Because of the importance of the theorem, let us give the proof of it:

Definition 6.10. The *time-shift operator* of order $n \in \mathbb{N}$ is defined on the environment as:

$$\theta_n : \begin{cases} \Omega & \longrightarrow \Omega \\ (\omega(k, y))_{(k, y) \in \mathbb{N} \times V} & \longmapsto (\omega(k + n, y))_{(k, y) \in \mathbb{N} \times V} \end{cases}$$

Its dual operator θ_n^* is defined on any function $f: \Omega \longrightarrow \mathbb{R}$ by $\theta_n^*(f(\omega)) := f(\theta_n \omega)$.

Lemma 6.11. Let $\beta \geq 0$, $x \in V$ and $n, m \in \mathbb{N}$. Then $(W_{k,x}^\beta)_{k \geq 0}$ satisfies the following recursion formula:

$$W_{n+m,x}^\beta = \sum_{y \in V} \widehat{W}_{n,x}^\beta(y) \theta_n^* W_{m,y}^\beta ,$$

where $\widehat{W}_{n,x}^\beta(y) = P_x [e^{\beta H_n(\omega, S) - n\lambda(\beta)} \mathbf{1}_{S_n=y}]$ is the point-to-point (normalized) partition function.

Proof. The formula follows from Markov's property:

$$\begin{aligned} W_{n+m,x}^\beta &= P_x \left[e^{\beta H_n(\omega, S) - n\lambda(\beta)} e^{\beta \sum_{i=n+1}^{n+m} \omega(i, S_i) - m\lambda(\beta)} \right] \\ &= \sum_{y \in V} P_x \left[e^{\beta H_n(\omega, S) - n\lambda(\beta)} \mathbf{1}_{S_n=y} e^{\beta \sum_{i=n+1}^{n+m} \omega(i, S_i) - m\lambda(\beta)} \right] \\ &= \sum_{y \in V} \widehat{W}_{n,x}^\beta(y) P_y \left[e^{\beta \sum_{i=n+1}^{n+m} \omega(i, S_i) - m\lambda(\beta)} \right] \quad (\text{Markov's property}) \\ &= \sum_{y \in V} \widehat{W}_{n,x}^\beta(y) \theta_n^* W_{m,y}^\beta . \end{aligned}$$

□

Remark 6.12. Notice that the sum on the right-hand side has a finite support given by $R_n(x)$ (this set is finite under (CLF) because of (5.2)), since for any $y \notin R_n(x)$, $P_x(S_n = y) = 0$. Conversely, if $y \in R_n(x)$, then $P_x(S_n = y) > 0$ by definition.

Proof of Theorem 6.9. Let us fix $\beta \geq 0$ and $x \in V$. Using Lemma 6.11, $Q \circ \theta_n^{-1} = Q$ and letting $m \rightarrow +\infty$, we obtain:

$$W_{\infty,x}^\beta = \sum_{y \in R_n(x)} \widehat{W}_{n,x}^\beta(y) \theta_n^* W_{\infty,y}^\beta \quad \text{a.s.}$$

Next, notice that for any $y \in R_n(x)$, $\widehat{W}_{n,x}^\beta(y) > 0$, so that:

$$\left\{ W_{\infty,x}^\beta = 0 \right\} = \left\{ \forall y \in R_n(x), \theta_n^* W_{\infty,y}^\beta = 0 \right\} ,$$

hence $\left\{ W_{\infty,x}^\beta = 0 \right\}$ is $\mathcal{G}_n$ -measurable with $\mathcal{G}_n := \sigma(\{\omega(k, y) \mid k \geq n, y \in V\})$ for any $n \in \mathbb{N}$. Therefore, it is a tail event, and Kolmogorov's zero-one law yields $W_{\infty,x}^\beta = 0$ a.s. or $W_{\infty,x}^\beta > 0$ a.s.

Moreover, if $W_{\infty,x}^\beta = 0$ a.s., then for any $n \in \mathbb{N}$:

$$0 = Q \left[W_{\infty,x}^\beta \right] = \sum_{y \in R_n(x)} \underbrace{Q \left[\widehat{W}_{n,x}^\beta(y) \right]}_{>0} Q \left[\theta_n^* W_{\infty,y}^\beta \right] .$$

Hence, since the environment is i.i.d., we get, for any $y \in R_n(x)$, $Q \left[W_{\infty,y}^\beta \right] = Q \left[\theta_n^* W_{\infty,y}^\beta \right] = 0$. Then, since $W_{\infty,y}^\beta$ is nonnegative a.s., this yields $W_{\infty,y}^\beta = 0$ a.s. Eventually, since the random walk is irreducible, $V = \bigcup_{n \geq 0} R_n(x)$, thus:

$$W_{\infty,x}^\beta = 0 \text{ a.s.} \iff \forall y \in V \quad W_{\infty,y}^\beta = 0 \text{ a.s.} \quad (6.4)$$

This concludes the first part of the theorem. Regarding the existence of β_c , let us fix $\theta \in (0, 1)$. Since $Q[W_{n,x}^\beta] = 1$, $((W_{n,x}^\beta)^\theta)_{n \geq 0}$ is uniformly integrable from La Vallée-Poussin criterion (see Corollary A.2 in

Appendix A). Furthermore, it converges to $(W_{\infty,x}^\beta)^\theta$ a.s., thus Lebesgue-Vitali's theorem yields the convergence in L^1 (see Proposition A.3 in Appendix A). Now, notice that for any $n \geq 0$, $\beta \mapsto Q[(W_{n,x}^\beta)^\theta]$ is non-increasing. Indeed, setting $e_n := e^{\beta H_n(\omega, S) - n\lambda(\beta)}$, we get:

$$\begin{aligned} \frac{d}{d\beta} Q[(W_{n,x}^\beta)^\theta] &= \theta Q \left[(W_{n,x}^\beta)^{\theta-1} \frac{d}{d\beta} W_{n,x}^\beta \right] \\ &= \theta P_x \left[Q \left[(W_{n,x}^\beta)^{\theta-1} e_n (H_n(\omega, S) - n\lambda'(\beta)) \right] \right] \quad (\text{Fubini}) \\ &= \theta P_x \left[\tilde{Q} \left[(W_{n,x}^\beta)^{\theta-1} (H_n(\omega, S) - n\lambda'(\beta)) \right] \right], \end{aligned}$$

where $d\tilde{Q} := e_n dQ$ (notice that $Q[e_n] = 1$). The environment is still independent under the new probability measure $\tilde{Q}$: therefore, using the FKG inequality (see Proposition A.5 in Appendix A) with $H_n(\omega, S)$ increasing with respect to the environment (for the coordinate order) and $(W_{n,x}^\beta)^{\theta-1}$ non-increasing yields

$$\begin{aligned} \frac{d}{d\beta} Q[(W_{n,x}^\beta)^\theta] &\leq \theta P_x \left[\tilde{Q} \left[(W_{n,x}^\beta)^{\theta-1} \right] \tilde{Q} [H_n(\omega, S) - n\lambda'(\beta)] \right] \\ &= \theta P_x \left[\tilde{Q} \left[(W_{n,x}^\beta)^{\theta-1} \right] \frac{d}{d\beta} Q[e_n] \right] = 0, \end{aligned}$$

since $Q[e_n] = 1$ is constant in β . Hence, $\beta \mapsto Q[(W_{n,x}^\beta)^\theta]$ is non-increasing, and using the convergence in L^1 , we obtain that $\beta \mapsto Q[(W_{\infty,x}^\beta)^\theta]$ is non-increasing as well. Then, setting

$$\beta_c(x) := \inf \left\{ \beta \geq 0 \mid Q[(W_{\infty,x}^\beta)^\theta] = 0 \right\},$$

we can check (on the basis of (6.4)) that $\beta_c(x) = \beta_c(y)$ for any $y \in V$. Indeed, if $\beta_c(x) = +\infty$, then $W_{\infty,x}^\beta > 0$ a.s. for any $\beta \geq 0$, and from (6.4) we obtain that $W_{\infty,y}^\beta > 0$ a.s. for any $\beta \geq 0$ as well, thus $\beta_c(y) = +\infty$. If $\beta_c(x) < +\infty$, there exists $\beta \geq 0$ such that $W_{\infty,x}^\beta = 0$ a.s., and once again (6.4) yields $W_{\infty,y}^\beta = 0$ a.s. so that $\beta_c(y) \leq \beta_c(x) < +\infty$. Exchanging the roles of x and y also yields $\beta_c(y) \leq \beta_c(x)$. Finally, $\beta_c := \beta_c(x)$ clearly separates the weak and strong disorder regimes. $\square$

We conclude this section by stating the main problems surrounding this phase transition, which will guide the state of the art later presented. The problems are the following:

- Find sufficient conditions for $\beta_c \in \{0, +\infty\}$, in which case there is no true phase transition, or for $\beta_c \in (0, +\infty)$, in which case there is indeed a phase transition between weak and strong disorder.
- State a distinction between weak and strong disorder which proves that these phases correspond to diffusive or localized behaviour of the polymer.
- Understand the behaviour of the polymer at criticality, i.e. when $\beta = \beta_c$.

In the case where $\beta_c = +\infty$, the system is never remarkably influenced by its environment regardless of its intensity. This situation is called *disorder irrelevance*, and should mean that an obstacle can always be avoided or that obstacles are rare because of the distribution of the environment. As we only consider weighted graphs that satisfy (CLF), the first possibility is unexpected, and the second possibility should be the exception rather than the rule.

The case $\beta_c = 0$ means that the slightest influence of the environment is seriously affecting the polymer. Following our expectations mentioned earlier, this could happen for the SRW in $\mathbb{Z}^d$ if d is small. A similar belief should apply to simple random walks on general weighted graphs, for a notion of dimension later defined.

In contrast, β_c should be finite and positive for random walks on weighted graphs with high dimension.

7 State of the art

In this section, we present the state of the art on the *Directed Polymer in Random Environments (DPRE)* model, primarily focusing on the questions mentioned earlier. While we present it for the model defined on general weighted graphs, as we introduced it, we will highlight many results concerning the most studied case, namely the case of the SRW in $\mathbb{Z}^d$ ($(G, \mu) = (\mathbb{Z}^d, \mathbb{1}_{x \sim y})$). This particular case has been studied from the origins of the model [HH85] to the very recent works of Stefan Junk and Hubert Lacoin [Jun21, JL24, JL25, Jun25, JL26], through the works of Bolthausen, Carmona, Hu, Yoshida, Shiga, Comets [Bol89, CH02, CSY03, CY06] and many others. We will also emphasize some results that have been proved for general weighted graphs, either by Clément Cosco, Inbar Seroussi and Ofer Zeitouni in [CSZ21], or by Naotaka Kajino, Kosei Konishi and Makoto Nakashima in [KKN20].

7.1 Non-triviality of the phase transition

In the previous section, we explained the natural question that arise regarding the model knowing Theorem 6.9. The first consisted in establishing whether the phase transition is trivial ($\beta_c \in \{0, +\infty\}$) or not ($\beta_c \in (0, +\infty)$). We also gave some motivations for why the case $\beta_c = +\infty$ should hold only for very particular random environments, while the case $\beta_c = 0$ should hold for low dimensions (a notion that still needs to be defined in all generality) and $\beta_c \in (0, +\infty)$ should hold for large dimensions. For the SRW in $\mathbb{Z}^d$, these intuitions have actually been proven, using other phase transitions for the model.

On the one hand, as it is often convenient to practice L^2 computations on stochastic processes, one could be interested in the so-called L^2 -region:

$$\mathbf{L2} := \left\{ \beta \geq 0 \mid \sup_{n \geq 0} Q \left[(W_{n,x}^\beta)^2 \right] < +\infty \right\}. \quad (7.1)$$

Remark 7.1. $\sup_{n \geq 0} Q \left[(W_{n,x}^\beta)^2 \right] = \lim_{n \geq 0} Q \left[(W_{n,x}^\beta)^2 \right]$ since $((W_{n,x}^\beta)^2)_{n \geq 0}$ is a submartingale.

Interestingly, for the DPRE model, the L^2 -region is characterized by another phase transition:

Proposition 7.2 ([CSZ21, Proposition 1.11]). *There exists a critical L^2 -threshold $\beta_2 \in [0, +\infty]$ such that for any $x \in V$:*

$$\begin{cases} \sup_{n \geq 0} Q[(W_{n,x}^\beta)^2] < +\infty & \text{if } \beta < \beta_2. \\ \sup_{n \geq 0} Q[(W_{n,x}^\beta)^2] = +\infty & \text{if } \beta > \beta_2. \end{cases}$$

The critical threshold β_2 is independent of the starting point $x \in V$, and the following inequality holds:

$$\beta_2 \leq \beta_c. \quad (7.2)$$

Indeed, for $\beta < \beta_2$, the L^2 -boundedness of $(W_{n,x}^\beta)_{n \geq 0}$ yields $Q[\sup_{n \geq 0} W_{n,x}^\beta] < +\infty$ from Doob's martingale inequality in L^2 . Then, from the dominated convergence theorem, $Q[W_{\infty,x}^\beta] = \lim_{n \rightarrow +\infty} Q[W_{n,x}^\beta] = 1$, so that $W_{\infty,x}^\beta$ cannot be null a.s. Hence weak disorder holds and $\beta < \beta_c$.

On the other hand, in statistical mechanics (or thermodynamics), the *free energy* of a system is naturally defined as the function whose local minima correspond to the equilibrium states of this system. In our case, this function is defined as:

Definition 7.3. Let $x \in V$. The (*normalized*) *free energy* of the system is defined as:

$$\mathfrak{f}_x(\beta) := \limsup_{n \rightarrow \infty} \frac{1}{n} Q \left[\log W_{n,x}^\beta \right].$$

Remark 7.4. Since $W_{n,x}^\beta = Z_{n,x}^\beta / Q[Z_{n,x}^\beta]$, Jensen's inequality yields $f_x(\beta) \leq 0$.

Free energy would be more naturally defined as a simple limit, but the existence of this limit is not given in all generality. Nonetheless, sufficient conditions on the weighted graph have been given in [KKN20, Proposition 2.1] for the limit to exist, and include the case of the SRW in $\mathbb{Z}^d$. In all generality, Definition 7.3 is sufficient to state the existence of another phase transition:

Proposition 7.5 ([CSZ21, Proposition 1.8]). $f_x(\beta) =: f(\beta)$ does not depend on x . Also, $f(\beta) = \limsup_{n \rightarrow +\infty} \frac{1}{n} \log W_{n,x}^\beta$ a.s. Furthermore, there exists $\bar{\beta}_c \in [0, +\infty]$ such that for any $x \in V$:

$$\begin{cases} f(\beta) = 0 & \text{for } \beta < \bar{\beta}_c, \\ f(\beta) < 0 & \text{for } \beta > \bar{\beta}_c \end{cases} \quad (\text{Very Strong Disorder (VSD)}) .$$

This proposition shows that for $\beta > \bar{\beta}_c$ so that $f(\beta) < 0$, not only does $W_{n,x}^\beta$ tend to zero (which is the definition of strong disorder), but it does so exponentially fast. This motivates the terminology *Very Strong Disorder (VSD)* for the set $\{\beta \geq 0 \mid f(\beta) < 0\}$. Of course, this implies that:

$$\beta_c \leq \bar{\beta}_c . \quad (7.3)$$

7.1.1 The case of the simple random walk in the lattice

In this part, let us consider the case $(G, \mu) = (\mathbb{Z}^d, \mathbb{1}_{x \sim y})$ of the SRW in the lattice $\mathbb{Z}^d$. In this case, the critical parameters β_2 , β_c and $\bar{\beta}_c$ are known to be all null for $d = 1, 2$ and all positive for $d \geq 3$. This dichotomy has been proved for β_c by [CH02] and [CSY03] for $d = 1, 2$ and [CY06] for $d \geq 3$. However, given (7.2) and (7.3), it is sufficient to prove:

$$\bar{\beta}_c = 0 \text{ for } d = 1, 2 \quad \text{and} \quad \beta_2 > 0 \text{ for } d \geq 3$$

to conclude on all critical parameters.

Definition 7.6. Let $(S_n)_{n \geq 0}$ be the SRW in $\mathbb{Z}^d$, starting at 0. We define:

$$\pi_d := P_0(\exists n \geq 1, S_n = 0) .$$

Recall from Polya's theorem that $(S_n)_{n \geq 0}$ is recurrent for $d = 1, 2$, hence $\pi_d = 1$, and transient for $d \geq 3$ so that $\pi_d < 1$. A consequence of ([Bol89], Theorem 2) gives an explicit formula for β_2 , using π_d , that yields $\beta_2 > 0$ when $d \geq 3$:

Theorem 7.7. Suppose that $d \geq 3$. Recall the definition of L2 from (7.1). Then, for any $\beta \geq 0$:

$$\beta \in \text{L2} \iff \lambda(2\beta) - 2\lambda(\beta) < \ln(1/\pi_d) . \quad (7.4)$$

Corollary 7.8. If $d \geq 3$, then $\beta_2 > 0$.

Proof. $\Lambda(\beta) := \lambda(2\beta) - 2\lambda(\beta)$ is continuous and $\Lambda(0) = 0$, so that the right-hand side of (7.4) holds for $\beta \geq 0$ small enough when $\pi_d < 1$, regardless of the distribution of the environment. $\square$

On the other hand, it has been shown in [Lac10] that $\bar{\beta}_c = 0$ for $d = 1, 2$:

Theorem 7.9 ([Lac10, Theorems 1.4 and 1.6]). There exists some constants $c > 0$ and $\beta_0 > 0$ (depending on the environment) such that for all $\beta \leq \beta_0$:

$$\begin{cases} f(\beta) \leq -c\beta^4 & \text{for } d = 1, \\ f(\beta) \leq -\exp(-c/\beta^4) & \text{for } d = 2. \end{cases}$$

In particular, $\bar{\beta}_c = 0$ for $d = 1, 2$.

Hence, the dichotomy is proven. Some refinements can be given to compare the values of the different critical parameters. In particular, it has been shown that $\beta_2 < \beta_c$ for $d \geq 4$ in [BS10] and for $d = 3$ in [BT10, JL24].

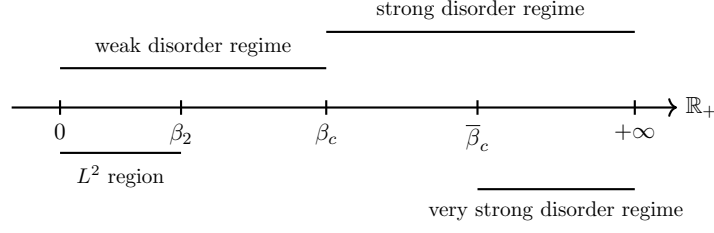


Figure 1: Phase transitions of the directed polymer on $(\mathbb{Z}^d, \mathbf{1}_{x \sim y})$, $d \geq 3$.

More recently, Stefan Junk and Hubert Lacoin have proved that $\beta_c = \bar{\beta}_c$ and $\beta_c \in \mathbf{WD}$ for $d \geq 3$ [JL26]. Their results in [JL25, JL26] will be further discussed in Section 7.3.

7.1.2 For general weighted graphs

In this part, we go back to the general setting of a weighted graph (G, μ) , and the naturally defined random walk $(S_n)_{n \geq 0}$ on (G, μ) , starting at a vertex $x \in V$.

Establishing conditions for β_c to be positive on general weighted graphs has been the objective of [CSZ21, KKN20]. For the SRW in $\mathbb{Z}^d$, a dichotomy appears between the case $d = 1, 2$ and the case $d \geq 3$. As mentioned above, this dichotomy corresponds to a change of behaviour for simple random walks in $\mathbb{Z}^d$, which are recurrent for $d = 1, 2$ and transient for $d \geq 3$. Therefore, one could investigate the way in which the recurrence or transience of the random walk affects the positivity of the critical inverse temperature. Such results are given in [CSZ21].

Theorem 7.10 ([CSZ21, Theorem 2.1]). *If $(S_n)_{n \geq 0}$ is recurrent and if $\inf_{y \in V} \mu_y > 0$, then $\beta_2 = 0$.*

Remark 7.11. In particular, if $\mu_{y,z} = \mathbf{1}_{y \sim z}$ so that $(S_n)_{n \geq 0}$ is the SRW in G , then $\inf_{y \in V} \mu_y = \inf_{y \in V} \#N(y) \geq 1$ since G is assumed to be connected.

Theorem 7.12 ([CSZ21, Theorem 2.11]). *If $(S_n)_{n \geq 0}$ is positive recurrent, then $\beta_c = 0$.*

Hence, there exists an apparent connection between the recurrence of the random walk and a trivial weak disorder regime. However, assuming only that $(S_n)_{n \geq 0}$ is recurrent is not sufficient to conclude that $\beta_c = 0$.

Conversely, transient walks with additional assumptions tend to possess a non-trivial weak disorder region:

Definition 7.13. Let (G, μ) be a weighted graph. The conductance μ is said to satisfy the *bounded weights* assumption if there exists $C_\mu > 0$ such that:

$$\forall x, y \in V \quad C_\mu^{-1} \leq \mu_{x,y} \leq C_\mu. \quad (\text{BW})$$

Theorem 7.14 ([CSZ21, Theorem 2.6]). *Assume that $(S_n)_{n \geq 0}$ is transient and that μ satisfies (BW). Furthermore, assume that*

$$\sup_{y \in V} G(y, y) < +\infty$$

where $G(y, z) := \sum_{k \geq 0} P_y(S_k = z)$ is the Green function. Then $\beta_2 > 0$.

Remark 7.15. Nonetheless, there exist transient random walks for which $\beta_c = 0$. An example of this is given by $G = \mathbb{N}$ and $\mu_{i,i+1} = e^{e^i}$, as mentioned in [CSZ21, Remark 2.10].

It would seem that the distinction between recurrent and transient random walks might not be the only factor for the existence of a non-trivial weak disorder regime. Another approach taken by [CSZ21, KKN20]

has been to consider a restricted (but quite large) class of graphs, namely those with convenient bounds on the heat kernel $p_n(x, y) := P_x(S_n = y)$.

SRWs on lattices and fractal graphs typically satisfy the following *Heat Kernel Estimates*. There exist some constants $c_1, c_2, c_3, c_4 > 0$ and some exponents $d_f > 0$, $d_w > 1$ such that for any $n \in \mathbb{N}^*$ and $x, y \in V$ such that $d(x, y) \leq n$:

$$p_n(x, y) \leq \frac{c_1}{n^{d_f/d_w}} \exp\left(-c_2 \left(\frac{d(x, y)^{d_w}}{n}\right)^{\frac{1}{d_w-1}}\right), \quad (\text{UHK})$$

$$p_n(x, y) + p_{n+1}(x, y) \geq \frac{c_3}{n^{d_f/d_w}} \exp\left(-c_4 \left(\frac{d(x, y)^{d_w}}{n}\right)^{\frac{1}{d_w-1}}\right). \quad (\text{LHK})$$

Example 7.16. The SRW in $\mathbb{Z}^d$ satisfies these estimates with $d_f = d$, $d_w = 2$. The SRW in the Sierpinski gasket graph does as well, with $d_f = \frac{\log 3}{\log 2}$ and $d_w = \frac{\log 5}{\log 2}$.

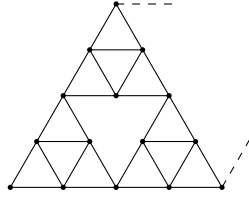


Figure 2: The Sierpinski gasket graph.

When these estimates are satisfied, the exponent $d_f > 0$ actually corresponds to the *Hausdorff dimension* of the graph, and the exponent $d_w > 1$ is called the *walk dimension* of the graph. The latter terminology comes from the physics literature and can be understood through the following heuristic:

$$P_x[\tau(x, r)] \asymp r^{d_w},$$

where $\tau(x, r)$ is the exit time of the ball $B(x, r)$, knowing that the random walk started in x . We then define the spectral dimension

$$d_s := \frac{2d_f}{d_w},$$

which appears naturally when trying to determine whether the random walk is recurrent or transient. Indeed, for any $x \in V$ and $n \in \mathbb{N}$,

$$p_n(x, x) \asymp n^{-\frac{d_s}{2}},$$

so that when $d_s < 2$, the series of general term $p_n(x, x)$ diverges at a polynomial growth rate. Random walks satisfying (UHK) and (LHK) with $d_s < 2$ are hence called strongly recurrent.

Although (UHK) and (LHK) might seem technical, they imply some natural hypotheses to make on the weighted graph (G, μ) . For instance, under (UHK) and (LHK), (BW) holds, as well as :

$$\sup_{x \in V} \#N(x) < +\infty, \quad (\text{BG})$$

$$\exists C_V > 0 \quad \forall x \in V \quad \forall r > 0 \quad C_V^{-1} \leq \mu_x \leq C_V \quad \text{and} \quad C_V^{-1} \nu(r) \leq V(x, r) \leq C_V \nu(r), \quad (\text{VG})$$

where $V(x, r) = \mu(B(x, r))$ and $\mu(\{x\}) := \mu_x$ is the conductance measure. (BG) means that G has a bounded geometry in the sense of the degrees of its vertices, and (VG) states the existence of a constant controlling G 's volume growth in the sense of its conductance.

Under these heat kernel estimates, the authors of [KKN20] have been able to demonstrate that strongly recurrent random walks lead to a trivial weak disorder regime :

Theorem 7.17 ([KKN20, Theorem 1.3]). • Assume (VG) and (UHK) with $d_s < 2$. Then there exists $C_U = C_U(\beta_0) > 0$ and $\beta_0 > 0$ such that for any $\beta \in [0, \beta_0]$,

$$\mathfrak{f}(\beta) \leq -C_U \beta^{\frac{4}{2-d_s}}.$$

In particular, $\overline{\beta_c} = 0$ (thus $\beta_2 = \beta_c = 0$).

- Suppose that, additionally, (LHK) holds. Then there exists $C_L = C_L(\beta_0) > 0$ such that for any $\beta \in [0, \beta_0]$,

$$\mathfrak{f}(\beta) \geq -C_L \beta^{\frac{4}{2-d_s}}.$$

Conversely, the authors of [CSZ21] have proved the existence of a non-trivial L^2 -region for "strongly transient" random walks:

Theorem 7.18 ([CSZ21, Theorem 2.14-ii]). *Assume that (S_n) satisfies (UHK) and (LHK) with $d_s > 2$. Then $\beta_2 > 0$ (hence $\bar{\beta}_c$ and β_c are also positive).*

7.2 A mathematical description of the polymer's behaviour in the different phases

In the introduction, we discussed some motivations for the terminology of weak and strong disorder. The polymer should behave approximately like the initial random walk in the weak disorder regime, whereas in the strong disorder regime, the environment should constrain the polymer to choose paths that are very different from those of the initial random walk with high probability. A rigorous mathematical description of such a difference in behaviour has been done in [CSY03] in the case of the SRW in $\mathbb{Z}^d$. More precisely, it has been shown that whenever β belongs to the weak disorder regime, the polymer has a diffusive behaviour that corresponds to that of the SRW in $\mathbb{Z}^d$. In contrast, they show a localized behaviour, distinguishing the polymer from the initial random walk, only in the very strong disorder regime ($\beta > \bar{\beta}_c$). This further motivates the analysis of the free energy function done in [KKN20], and the interest in comparing the strong and the very strong disorder regimes (i.e. establishing how much $\bar{\beta}_c$ is greater than β_c) as done in [JL26].

Proposition 7.19 ([CSY03, Proposition 2.4]). *Assume that (S_n) is the SRW in $\mathbb{Z}^d$. Then :*

- For $\beta \geq 0$ in the weak disorder regime, delocalization holds : $\max_{y \in \mathbb{Z}^d} P_{n-1,0}^\beta(S_n = y) \xrightarrow{n \rightarrow +\infty} 0$.
- For $d = 1, 2$ or for $\beta > \bar{\beta}_c$, localization holds (there exist favorite sites). For some $c = c_{d,\beta} > 0$,

$$\limsup_{n \rightarrow +\infty} \max_{y \in \mathbb{Z}^d} P_{n-1,0}^\beta(S_n = y) \geq c \quad \mathbb{P} - a.s.$$

Remark 7.20. The second part of the proposition has been extended on general weighted graphs (cf. Theorem 1.10 in [CSZ21]). This further motivates the study of $\mathfrak{f}$ conducted in [KKN20].

Still in the case of the SRW in $\mathbb{Z}^d$, an even more precise result has been given in [BC21] regarding the behaviour of the polymer in the very strong disorder regime. They showed that in the very strong disorder regime, so-called asymptotic pure atomicity holds : the mass of the endpoint is concentrated on small regions. Letting

$$\mathcal{A}_k^\varepsilon := \left\{ y \in \mathbb{Z}^d \mid P_{k,0}^\beta(S_k = x) > \varepsilon \right\},$$

they prove that for any sequence $(\varepsilon_k)_{k \geq 0}$ vanishing as $k \rightarrow +\infty$,

$$\lim_{n \rightarrow +\infty} \sum_{k=0}^{n-1} P_{k,0}^\beta(S_k \in \mathcal{A}_k^{\varepsilon_k}) = 1 \quad \mathbb{P} - a.s.$$

whenever β belongs to the very strong disorder regime.

Finally, to the purpose of comparing the strong and very strong disorder regimes, it is already known that $\beta_c = \bar{\beta}_c = 0$ for the SRW in $\mathbb{Z}^d$ with $d = 1, 2$. In a (very) recent breakthrough, Stefan Junk and Hubert Lacoin have proved the following :

Theorem 7.21 ([JL26, Theorem 2.1]). *Let $(S_n)_{n \geq 0}$ be the SRW in $\mathbb{Z}^d$ with $d \geq 3$ (trivial otherwise). Then $\beta_c = \bar{\beta}_c$.*

Hence, combining this result with Proposition 7.19 justifies the terminology behind weak and strong disorder as intended. The results of [JL25, JL26] leading to this equality will be further presented in Section 7.3, as they also lead to an answer to the behaviour of the polymer at criticality.

7.3 Understanding the polymer walk at criticality

In this subsection, we consider the SRW in $\mathbb{Z}^d$ once again. This far, we have seen that the directed polymer model experiences a phase transition at a certain $\beta_c \in [0, +\infty]$, and that for the SRW in $\mathbb{Z}^d$, $\beta_c > 0$ if, and only if $d \geq 3$. Furthermore, $\beta_c = \bar{\beta}_c$ and the phase transition consists of a change of behaviour between a diffusive one in the weak disorder regime and a localized behaviour in the (very) strong disorder regime. Finally, from the definition of β_c , $[0, \beta_c[\subseteq \mathbf{WD}$ and $]\beta_c, +\infty[\subseteq \mathbf{SD}$. One of the main remaining questions is to establish whether $\beta_c \in \mathbf{WD}$ or $\beta_c \in \mathbf{SD}$ (the proof of Theorem 6.9 excludes any other possibility). This question has first been treated in some particular choices for the environment (*e.g.* $\omega(1, 0) \sim \mathcal{N}(0, 1)$ or $\omega(1, 0) \sim \mathcal{B}(p)$ for some $p \in [0, 1]$) or additional hypotheses, such as assuming that the environment is bounded from above. Later, from 2024 to now on, Stefan Junk and Hubert Lacoin have been able to treat this problem for general environments in a fruitful series of articles [JL24, JL25, JL26]. Their works have been the principal object of study of this internship, in the hope of extending them to other weighted graphs. In this part, we present their main results.

Let us stress once again that in all that follows, $(G, \mu) = (\mathbb{Z}^d, (\mathbf{1}_{y \sim z})_{y, z \in \mathbb{Z}^d})$ so that $(S_n)_{n \geq 0}$ is just the SRW in $\mathbb{Z}^d$, starting at $x = 0$. For simplicity, we write $W_n^\beta := W_{n,0}^\beta$ and $W_\infty^\beta := W_{\infty,0}^\beta$. The approach of Junk and Lacoin has been to consider the moments of the limit partition function W_∞^β in order to establish some control on its distribution via its tails. For that purpose, let us define the principal object of this subsection, namely the *integrability parameter*

$$p^*(\beta) := \inf \left\{ p > 0 \mid \sup_{n \geq 0} Q \left[\left(W_n^\beta \right)^p \right] = +\infty \right\}.$$

Notice that we always have $p^*(\beta) \geq 1$. Indeed, $((W_n^\beta)^p)_{n \geq 0}$ is a supermartingale for $p < 1$, hence $Q[(W_n^\beta)^p] \leq Q[(W_0^\beta)^p] = 1$ for any $n \in \mathbb{N}$. When $p^*(\beta) > 1$, using Doob's martingale L^p -inequality, we get $Q \left[\sup_{n \geq 0} W_n^\beta \right] < +\infty$, and the dominated convergence theorem then yields $Q[W_\infty^\beta] = 1$ so that $\beta \in \mathbf{WD}$. More interestingly, the converse also holds under some assumption on the environment:

Theorem 7.22 ([Jun21, Theorem 1.1]). *Weak disorder implies $Q \left[\sup_{n \geq 0} W_n^\beta \right] < +\infty$. Furthermore, if $\text{ess sup } \omega(1, 0) < +\infty$, then weak disorder implies $p^*(\beta) > 1$.*

The results of [JL25] will then show that the assumption $\text{ess sup } \omega(1, 0) < +\infty$ can be dropped. Therefore, $p^*(\beta)$ is a distinguishing object between weak and strong disorder, as $p^*(\beta) > 1$ in $\mathbf{WD}$ and $p^*(\beta) = 1$ in $\mathbf{SD}$.

For a later study of the tails of W_∞^β , let us define :

$$\tau_A := \inf \left\{ n \geq 0 \mid W_n^\beta \geq A \right\} \tag{7.5}$$

the *overshooting time* of a threshold $A \geq 1$. The main reason for assuming an upper bounded environment is to get a control on $W_{\tau_A}^\beta$. Indeed, setting $L := e^{\beta \text{ess sup } \omega(1,0) - \lambda(\beta)}$, (7.5) yields, on the one hand, $W_{\tau_A}^\beta \geq A$, and on the other hand :

$$\begin{aligned} W_{\tau_A, x}^\beta &= P_x \left[\exp \left(\beta \sum_{i=1}^{\tau_A^x} \omega(i, S_i) - n\lambda(\beta) \right) \right] \\ &\leq L P_x \left[\exp \left(\beta \sum_{i=1}^{\tau_A^x - 1} \omega(i, S_i) - (n-1)\lambda(\beta) \right) \right] \\ &= L W_{\tau_A^x - 1, x}^\beta \\ &\leq LA \end{aligned}$$

on the event $\{\tau_A < +\infty\}$. Hence, $W_{\tau_A}^\beta$ is comparable to the threshold A when the environment is bounded from above. Without any assumption on the environment, Junk and Lacoïn have then proved a similar control :

Proposition 7.23 ([JL25, Proposition 2.3]). *For any $p \geq 1$, there exists $C_p = C_p(\omega, \beta) > 1$ such that for any $A \geq 1$:*

$$Q \left[\left(W_{\tau_A}^\beta \right)^p \mid \tau_A < +\infty \right] \leq C_p A^p .$$

We prove this proposition in the context of general weighted graphs in Section 8.2. This proposition, along with Theorem 7.22, suffices to conclude on the equivalence between $\beta \in \mathbf{WD}$ and $p^*(\beta) > 1$:

Corollary 7.24 ([JL25, Corollary 2.8-i]). *Assume that weak disorder holds at $\beta \geq 0$. Then $p^*(\beta) > 1$.*

Proposition 7.23 also yields an estimate for the probability of the event $\{\tau_A < +\infty\}$ in which it holds.

Corollary 7.25 ([JL25, Corollary 2.6]). *There exists a constant $c = c(\beta, Q) > 0$ such that for any threshold $A \geq 1$:*

$$Q(\tau_A < +\infty) \geq c A^{-p^*(\beta)} .$$

Setting $W_{\infty}^{\beta,*} := \sup_{n \geq 0} W_n^\beta$ so that $Q(\tau_A < +\infty) = Q(W_{\infty}^{\beta,*} \geq A)$, the previous corollary gives an estimate on the tail of the supremum of the partition function. Similarly, setting

$$\widehat{W}_{\infty}^{\beta,*} := \sup_{(n,y) \in \mathbb{N} \times \mathbb{Z}^d} \widehat{W}_n^\beta(y)$$

to be the supremum of the point-to-point partition function, it can be shown that the tails of W_{∞}^β , $W_{\infty}^{\beta,*}$ and $\widehat{W}_{\infty}^{\beta,*}$ all decay like the power function of order $-p^*(\beta)$ in the weak disorder regime:

Theorem 7.26 ([JL25, Theorem 2.1]). *Assume that weak disorder holds at β . Then:*

$$Q(W_{\infty}^\beta > u) \asymp Q(W_{\infty}^{\beta,*} > u) \asymp Q(\widehat{W}_{\infty}^{\beta,*} > u) \asymp u^{-p^*(\beta)} .$$

This theorem depends on Proposition 7.23 and is the key to Junk and Lacoïn's results regarding the behavior of the polymer at criticality, and the equivalence between strong and very strong disorder mentioned in the last section. In particular, additional information on the map $\beta \mapsto p^*(\beta)$ follows from its link with the partition function (and its suprema associates). First, it is (at least) left-continuous in the weak disorder phase:

Corollary 7.27 ([JL25, Corollary 2.11-i]). *Assume that weak disorder holds at β . Then, $\lim_{\beta' \uparrow \beta} p^*(\beta') = p^*(\beta)$.*

In particular, $\beta \mapsto p^(\beta)$ is left-continuous on $(0, \beta_c)$.*

It has also been shown in [Jun25] that $\beta \mapsto p^*(\beta)$ is continuous for β in the weak disorder regime beyond the L^2 -phase, that is $\mathbf{WD} \setminus \mathbf{L2} \supseteq]\beta_2, \beta_c[$. Then, Theorem 7.26 gives an improved lower bound for $p^*(\beta)$ in the weak disorder regime:

Corollary 7.28 ([JL25, Corollary 2.8-ii]). *Assume that weak disorder holds at β . Then:*

$$p^*(\beta) \geq 1 + \frac{2}{d} > 1 .$$

In the end, Corollary 7.32 will show that this lower bound is sharp in general, computing that $p^*(\beta_c) = 1 + 2/d$. Thus, p^* characterizes the phase transition between the weak disorder phase, where it is left continuous and lower-bounded by a constant greater than one which depends only on the dimension, and the strong disorder phase, where it is constant and equal to one.

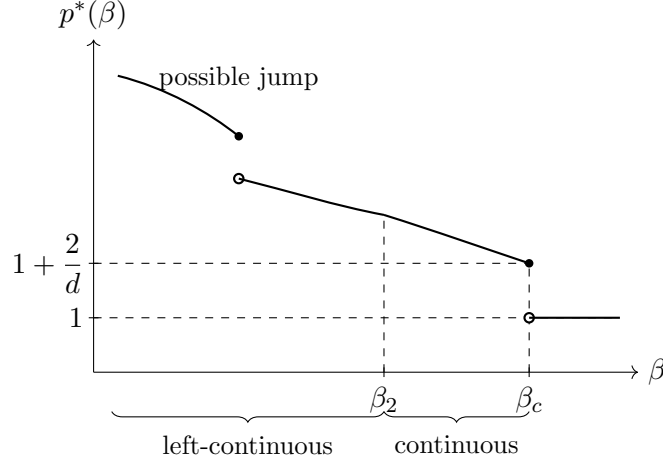


Figure 3: Illustration of the graph of $p^*(\beta)$

Let us now get back to the question of the previous section, that is mathematically describe the difference between weak and strong disorder. As said before, the distinction is clearer between weak and very strong disorder, hence the importance of Junk and Lacoïn's result in [JL26] stating that $\beta_c = \overline{\beta}_c$. To prove this statement, let us introduce

$$q^*(\beta) := \inf \left\{ p \geq 0 \mid \lim_{n \rightarrow \infty} \frac{1}{n} \log Q \left[\left(W_n^\beta \right)^p \right] > 0 \right\} .$$

The limit considered is well defined, since $(\log Q[(W_n^\beta)^p])_{n \geq 0}$ is superadditive or subadditive depending on whether $p \leq 1$ or $p \geq 1$, from Lemma 6.11, Jensen's inequality and invariance under translation of $\mathbb{Z}^d$ (equipped with its natural conductance). $q^*(\beta)$ is the smallest exponent $p \geq 0$ such that $(W_n^\beta)_{n \geq 0}$ diverges exponentially fast to infinity in L^p . In particular, we have

$$p^*(\beta) \leq q^*(\beta) .$$

Also, it is possible to show that $q^*(\beta) = 1$ in the very strong disorder regime (following the results of [LW09], Section 6). In particular, $q^*(\beta) = p^*(\beta) (= 1)$ in the very strong disorder regime, and Theorem 7.26 also yields this equation in the weak disorder regime:

Corollary 7.29 ([JL25, Corollary 2.11-ii]). *Assume that weak disorder holds at $\beta \geq 0$. Then $p^*(\beta) = q^*(\beta)$.*

As mentioned in the previous section, as Proposition 7.23 and Theorem 7.26 ultimately lead to $\beta_c = \overline{\beta}_c$ in [JL26, Theorem 2.1], we finally obtain

$$\forall \beta \geq 0 \quad p^*(\beta) = q^*(\beta) .$$

Thus, the sequence $(W_n^\beta)_{n \geq 0}$ experiences a phase transition between the weak disorder regime, where it is L^p -bounded for some $p > 1$, and the (very) strong disorder regime, where it diverges exponentially fast in L^p for any $p > 1$. That constitutes another appreciable difference between both phases.

Finally, let us investigate the limit of $(W_n^\beta)_{n \geq 0}$ in $L^{p^*(\beta)}$. From Theorem 7.26, it holds that:

$$Q \left(\left(W_\infty^\beta \right)^{p^*(\beta)} > u \right) = Q \left(W_\infty^\beta > u^{\frac{1}{p^*(\beta)}} \right) \asymp u^{-1} ,$$

therefore:

$$\begin{aligned} \liminf_{n \rightarrow +\infty} Q \left[\left(W_n^\beta \right)^{p^*(\beta)} \right] &\geq Q \left[\liminf_{n \rightarrow +\infty} \left(W_n^\beta \right)^{p^*(\beta)} \right] && \text{(Fatou's lemma)} \\ &= Q \left[\left(W_\infty^\beta \right)^{p^*(\beta)} \right] \\ &\asymp \int_0^\infty \frac{1}{u} du \\ &= +\infty . \end{aligned}$$

Hence, $\left(Q \left[\left(W_n^\beta\right)^p \right] \right)_{n \geq 0}$ goes to infinity in $L^{p^*(\beta)}$. However, compared to the case $p > p^*(\beta)$, here the growth is sublinear:

Corollary 7.30 ([JL25, Corollary 2.14]). *Assume that weak disorder holds. Then, there exists $C = C_\beta > 0$ such that for any $n \in \mathbb{N}$:*

$$Q \left[\left(W_n^\beta\right)^{p^*(\beta)} \right] \leq Cn .$$

Finally, while Proposition 7.23 and Theorem 7.26 have been helpful to appreciate the distinctions between the different phases through $\beta \mapsto p^*(\beta)$, they are also the key tools for proving $\beta_c \in \mathbf{WD}$:

Theorem 7.31 ([JL26, Theorem 2.1]). *Let $(S_n)_{n \geq 0}$ be the SRW in $\mathbb{Z}^d$ (with $d \geq 3$). Then weak disorder holds at $\beta = \beta_c$.*

Corollary 7.32. *We can compute that $p^*(\beta_c) = 1 + \frac{2}{d}$, so that the lower bound of Corollary 7.28 is indeed sharp.*

Proof. Since weak disorder holds at β_c , Corollary 7.28 yields $p^*(\beta_c) \geq 1 + \frac{2}{d}$. Recall from [Jun25] that $\beta \mapsto p^*(\beta)$ is right-continuous at points $\beta \geq 0$ in the weak disorder phase beyond the L^2 -region, that are points such that $p^*(\beta) \in (1 + \frac{2}{d}, 2)$. Thus, since p^* is not right-continuous at β_c and $\beta_2 < \beta_c$ (cf. the end of Section 7.1.1), necessarily $p^*(\beta_c) \leq 1 + \frac{2}{d}$. Hence the result. $\square$

8 Open questions and personal developments

In the state of the art, we have given many results that give some answers to the principal questions regarding the initial model, namely the directed polymer associated to a simple random walk in $\mathbb{Z}^d$. Future works on the DPRE model in $(\mathbb{Z}^d, (\mathbf{1}_{y \sim z})_{y, z \in \mathbb{Z}^d})$, if not interested mainly in the stochastic heat equation, should for instance focus on the following conjectures:

Conjecture 8.1. *For any $k \geq 2$, there exists a critical parameter $\beta_k \in [0, +\infty]$ such that*

$$\begin{cases} \sup_{n \geq 0} Q[(W_n^\beta)^k] < +\infty & \text{if } \beta < \beta_k , \\ \sup_{n \geq 0} Q[(W_n^\beta)^k] = +\infty & \text{if } \beta > \beta_k . \end{cases}$$

Furthermore, it holds that

$$\beta_2 > \beta_3 > \dots > \beta_k > \dots .$$

Conjecture 8.2. *For any $\beta \geq 0$, $p^*(\beta)$ is finite.*

Another question surrounds the regularity of $\beta \mapsto f(\beta)$ at $\beta = \beta_c$. From Theorem 7.21, f is not analytical at the critical temperature, but Junk and Lacoïn have made some conjectures about its smoothness (see Conjecture 2.16 in [JL25]).

Our objective has been elsewhere and consisted of pursuing the way paved by [CSZ21, KKN20] of studying the DPRE model on general weighted graphs. More precisely, we have been interesting in extending the results of [JL25, JL26] to other weighted graphs.

Many of these results are based on general martingale analysis and inequalities using the recursion formula from Lemma 6.11 :

$$\forall \beta \geq 0 \quad \forall n, m \in \mathbb{N} \quad W_{n+m}^\beta = \sum_{y \in \mathbb{Z}^d} \widehat{W}_n^\beta(y) \theta_n^* W_{m,y}^\beta .$$

Often, such inequalities appear inside an expectation. Since:

1. the environment is i.i.d., we have $\theta_n^* W_{m,y}^\beta \stackrel{d}{\sim} W_{m,y}^\beta$ for any $y \in \mathbb{Z}^d$.

2. $(\mathbb{Z}^d, (\mathbb{1}_{y \sim z})_{y,z \in \mathbb{Z}^d})$ is invariant under translation (the group of translations acts transitively on it), we have $W_{m,y}^\beta \stackrel{d}{\sim} W_{m,z}^\beta$ for any $y, z \in \mathbb{Z}^d$.

Therefore, $\theta_n^* W_{m,y}^\beta$ can be moved from the sum and simply replaced by W_m^β (in distribution). Furthermore, the remaining term $\widehat{W}_n^\beta(y)$ is independent (in distribution) of the starting point.

In the broader context of general weighted graphs, the recursion formula still holds with $\widehat{W}_{n,x}^\beta(y)$ depending on the starting point $x \in V$. Since the environment is still assumed to be i.i.d., the first point above is still valid. However, two different starting points $x, x' \in V$ can lead to significantly different distributions for the sequences $(W_{n,x}^\beta)_{n \geq 0}$ and $(W_{n,x'}^\beta)_{n \geq 0}$. Consequently, adapting Junk and Lacoïn's arguments to general weighted graphs requires an analysis of the entire family of martingale sequences

$$\left((W_{n,x}^\beta)_{n \geq 0} \right)_{x \in V} ,$$

which can prove very hard to do in all generality. Junk and Lacoïn themselves have been interested in extending their results to other random walks in $\mathbb{Z}^d$, and have restricted themselves to random walks with i.i.d. increments.

Definition 8.3. A random walk $(S_n)_{n \geq 0}$ on $\mathbb{Z}^d$ (starting at 0) is said to have i.i.d. increments if there exists a probability measure ν on $(\mathbb{Z}^d, \mathcal{P}(\mathbb{Z}^d))$ such that :

$$\forall n \in \mathbb{N} \quad \forall y, z \in \mathbb{Z}^d \quad P_0(S_{n+1} = z \mid S_n = y) = \nu(z - y) ,$$

or equivalently if the sequence $(S_{n+1} - S_n)_{n \geq 0}$ is i.i.d. with $S_0 = 0$ and

$$\forall y \in \mathbb{Z}^d \quad P_0(S_1 = y) = \nu(y) .$$

In $\mathbb{Z}^d$, such random walks whose transition probabilities depend only on the end-point are quite natural to consider, as the lattice is invariant under translation. In contrast, for general weighted graphs, such random walks cannot be defined, as the local structure of the graph at one vertex can be very different from that at another.

In the following subsections, we present extensions of the results in [JL25] to a restrictive class of weighted graphs, namely quasi-transitive graphs. Such graphs constitute a natural generalization, keeping in mind our objective of comparing the partition function sequence for different starting points. An exception is Proposition 7.23, which can actually be extended to a broader class of weighted graphs, requiring only the condition (BW).

Definition 8.4. A *weighted rooted graph* (G, μ, v) is a collection of a graph $G = (V, E)$, a family of weights (conductance) $\mu = (\mu_{x,y})_{x,y \in V}$ on G , and a distinguished vertex $v \in V$ called the *root*.

Two weighted rooted graphs (G, μ, v) , (G', μ', v') are said to be *isomorphic*, denoted $(G, \mu, v) \cong (G', \mu', v')$, if there exists a bijection $\pi : V \rightarrow V'$ preserving the weighted graph structure at the corresponding roots, i.e. :

$$\pi(v) = v', \quad x \sim y \iff \pi(x) \sim \pi(y), \quad \mu_{\pi(x), \pi(y)} = \mu_{x,y} .$$

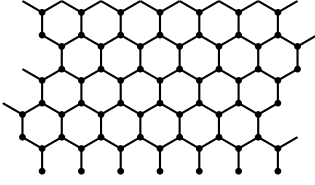
Hence, π preserves the root, edges and conductance.

Definition 8.5. A weighted graph (G, μ) is said to be *quasi-transitive* if there exists a finite set $V_0 \subseteq V$ such that for any $x \in V$, there exists $v_0 \in V_0$ such that $(G, \mu, x) \cong (G, \mu, v_0)$.

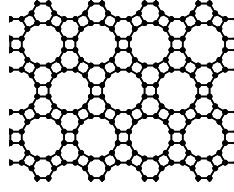
When $\#V_0 = 1$, (G, μ) is said to be *transitive*.

A graph G is said to be quasi-transitive if $(G, (\mathbb{1}_{y \sim z})_{y,z \in V})$ is.

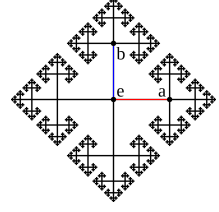
Transitive graphs include, for example, the lattice $\mathbb{Z}^d$ or the hexagonal lattice $\mathbb{H}$. More generally, any periodic tiling is quasi-transitive. One particular type of fractals are transitive, those are Cayley graphs of groups. Indeed, a group acts transitively on its Cayley graph by left-multiplication. However, the class of quasi-transitive graphs includes neither aperiodic tilings, like the Penrose tiling, nor typical fractals, like the Sierpinski gasket graph (Figure 2).



The hexagonal lattice $\mathbb{H}$ is transitive.



The great rhombitrihexagonal tiling is quasi-transitive.



The Cayley graph of the free group on two generators is transitive.

8.1 A characterization of weak disorder

Recall that, given any weighted graph (G, μ) , any starting point $x \in V$ and any inverse temperature $\beta \geq 0$, the following implications hold :

$$\begin{aligned} & \left(W_{n,x}^\beta \right)_{n \geq 0} \text{ is bounded in } L^p \text{ for some } p > 1, \text{ i.e. } \sup_{n \geq 0} Q \left[\left(W_{n,x}^\beta \right)^p \right] < +\infty . \\ \implies & \left(W_{n,x}^\beta \right)_{n \geq 0} \text{ is dominated, i.e. } Q \left[\sup_{n \geq 0} W_{n,x}^\beta \right] < +\infty . \\ \implies & \text{Weak disorder holds at } \beta . \end{aligned}$$

from Doob's L^p -martingale inequality and the dominated convergence theorem. The first step for proving Junk and Lacoïn's results was Theorem 7.22, stating that conversely, weak disorder implies domination. In all generality, this equivalence does not hold. Indeed, [CSZ21] gives an example of a weighted graph with a non-trivial weak disorder regime ($\beta_c > 0$), but with $\left(W_{n,x}^\beta \right)_{n \geq 0}$ never uniformly integrable (u.i) in the weak disorder region (see [CSZ21, Proposition 4.1]). In particular, domination does not hold for this weighted graph. Still, the authors have proved that, for quasi-transitive weighted graphs, weak disorder does imply uniform integrability (see [CSZ21, Proposition 2.23]). Adapting the proof of [JL25], we prove a stronger statement that is the generalization of Theorem 7.22 for quasi-transitive weighted graphs:

Lemma 8.6. *Let $\mu_{n,x}^\beta(y) := P_{n,x}^\beta(S_n = y)$. Then*

$$\widehat{W}_{n,x}^\beta(y) = \mu_{n,x}^\beta W_{n,x}^\beta .$$

Therefore, the recursion formula of Lemma 6.11 can be written as

$$W_{n+m,x}^\beta = W_{n,x}^\beta \sum_{y \in V} \mu_{n,x}^\beta(y) \theta_n^* W_{m,y}^\beta , \quad (8.1)$$

where the weights $(\mu_{n,x}^\beta(y))_{y \in V}$ are $\mathcal{F}_n$ -measurable and sum to 1, and $\theta_n^* W_{m,y}^\beta$ is independent from $\mathcal{F}_n$ and distributed as $W_{m,y}^\beta$ under Q .

Proposition 8.7. *Let (G, μ) be quasi-transitive. Then $Q \left[\sup_{n \geq 0} W_{n,x}^\beta \right] < +\infty$ in the whole weak disorder region.*

Proof. Our proof is directly inspired from the proof of ([Jun21], Theorem 1.1). Fix $x \in V$ and $\beta \geq 0$ such that weak disorder holds. Denote $W_{n,x} := W_{n,x}^\beta$ and :

$$W_{n,x}^* := \sup_{0 \leq k \leq n} W_{k,x} .$$

Let us prove the existence of $\varepsilon > 0$, $\eta > 0$ such that, for any $n \in \mathbb{N}$ and $t > 1$:

$$\eta Q(W_{n,x}^* > t) \leq Q(W_{n,x} > t\varepsilon) , \quad (8.2)$$

so that :

$$\begin{aligned}
Q[W_{n,x}^*] &= \int_0^\infty Q(W_{n,x}^* > t) \, dt \\
&= 1 + \int_1^\infty Q(W_{n,x}^* > t) \, dt \\
&\leq 1 + \frac{1}{\varepsilon\eta} \int_\varepsilon^\infty Q(W_{n,x} > t) \, dt \\
&\leq 1 + \frac{1}{\varepsilon\eta} \underbrace{Q[W_{n,x}]}_{=1} < +\infty ,
\end{aligned}$$

which will conclude, taking the limit as $n \rightarrow +\infty$ on the left-hand side. The second equation comes from the fact that $W_{0,x} = 1$ *a.s.*

Let us prove (8.2). Set, for $\varepsilon > 0$ to be chosen later :

$$f_\varepsilon(x) := \left(\frac{x}{\varepsilon} - 1\right) \wedge 1 ,$$

so that f_ε is increasing, concave, and satisfies the following inequalities for any $x \in \mathbb{R}$:

$$\mathbb{1}_{x \geq \varepsilon} \geq f_\varepsilon(x) \geq \mathbb{1}_{x \geq 2\varepsilon} - \mathbb{1}_{0 \leq x \leq \varepsilon} \quad (8.3)$$

Now, recall that for any $k, l \in \mathbb{N}$:

$$\frac{W_{k+l,x}}{W_{k,x}} = \sum_{y \in V} \mu_{k,x}(y) \theta_k^* W_{l,y} , \quad (8.4)$$

where the weights $(\mu_{k,x}(y))_{y \in V}$ are $\mathcal{F}_k$ -measurable and sum to 1, and $\theta_k^* W_{l,y}$ is independent from $\mathcal{F}_k$ and distributed as $W_{l,y}$ under Q .

Therefore, on the one hand, we have :

$$Q \left[f_\varepsilon \left(\frac{W_{k+l,x}}{W_{k,x}} \right) \mid \mathcal{F}_k \right] \geq Q \left[\sum_{y \in V} \mu_{k,x}(y) f_\varepsilon(\theta_k^* W_{l,y}) \mid \mathcal{F}_k \right] = \sum_{y \in V} \mu_{k,x}(y) Q[f_\varepsilon(W_{l,y})] ,$$

and on the other hand, (8.4) still holds if k is replaced by a stopping time τ on the event $\{\tau < +\infty\}$. Hence, setting $\tau := \min \{n \geq 0 \mid W_{n,x} > t\}$, we get :

$$\begin{aligned}
Q(W_{n,x} > t\varepsilon) &\geq Q \left(\tau \leq n, \frac{W_{n,x}}{W_{\tau,x}} > \varepsilon \right) \\
&= Q \left[\mathbb{1}_{\tau \leq n} \mathbb{1}_{\frac{W_{n,x}}{W_{\tau,x}} > \varepsilon} \right] \\
&\geq Q \left[\mathbb{1}_{\tau \leq n} f_\varepsilon \left(\frac{W_{n,x}}{W_{\tau,x}} \right) \right] \quad (\text{from (8.3)}) \\
&= Q \left[\mathbb{1}_{\tau \leq n} Q \left[f_\varepsilon \left(\frac{W_{n,x}}{W_{\tau,x}} \right) \mid \mathcal{F}_\tau \right] \right] \\
&\geq Q \left[\mathbb{1}_{\tau \leq n} \sum_{y \in V} \mu_{\tau,x}(y) Q[f_\varepsilon(W_{n-\tau,y})] \right] \\
&\geq \eta Q(\tau \leq n) = \eta Q(W_{n,x}^* > t) ,
\end{aligned}$$

where $\eta := \inf_{y \in V, k \geq 0} Q[f_\varepsilon(W_{k,y})]$.

All is left to show is that, for a good choice of $\varepsilon > 0$, we have $\eta > 0$. Notice that :

$$\begin{aligned}
\eta &= \inf_{y \in V, k \geq 0} Q[f_\varepsilon(W_{k,y})] \\
&\geq \inf_{y \in V} Q \left[\inf_{k \geq 0} f_\varepsilon(W_{k,y}) \right] && \text{(by positivity of the integral)} \\
&\geq \inf_{y \in V} Q \left[f_\varepsilon \left(\inf_{k \geq 0} W_{k,y} \right) \right] && \text{(f is increasing)} \\
&\geq \inf_{y \in V} \left(Q \left(\inf_{k \geq 0} W_{k,y} \geq 2\varepsilon \right) - Q \left(\inf_{k \geq 0} W_{k,y} \leq \varepsilon \right) \right) && \text{(from (8.3))}
\end{aligned}$$

Now, let any $y \in V$. By assumption, there exists $v_0 \in V_0$ such that $(G, y) \cong (G, v_0)$, and since the environment is *i.i.d* in time-space, $W_{k,y} \stackrel{d}{\sim} W_{k,v_0}$ for all $k \in \mathbb{N}$. It follows that $Q \left(\inf_{k \geq 0} W_{k,y} \geq 2\varepsilon \right) = Q \left(\inf_{k \geq 0} W_{k,v_0} \geq 2\varepsilon \right)$, therefore :

$$\eta \geq \min_{v_0 \in V_0} \left(Q \left(\inf_{k \geq 0} W_{k,v_0} \geq 2\varepsilon \right) - Q \left(\inf_{k \geq 0} W_{k,v_0} \leq \varepsilon \right) \right). \quad (8.5)$$

For any $v_0 \in V_0$, the monotone convergence theorem yields $Q \left(\inf_{k \geq 0} W_{k,v_0} \geq 2\varepsilon \right) \xrightarrow{\varepsilon \downarrow 0} Q \left(\inf_{k \geq 0} W_{k,v_0} > 0 \right)$ and $Q \left(\inf_{k \geq 0} W_{k,v_0} \leq \varepsilon \right) \xrightarrow{\varepsilon \downarrow 0} Q \left(\inf_{k \geq 0} W_{k,v_0} = 0 \right)$. Then, $\left\{ \inf_{k \geq 0} W_{k,v_0} > 0 \right\} = \{W_{\infty,v_0} > 0\}$ since 0 is an absorbing state for positive martingales. Therefore, since weak disorder holds independently of the starting point, $Q \left(\inf_{k \geq 0} W_{k,v_0} > 0 \right) = 1 = 1 - Q \left(\inf_{k \geq 0} W_{k,v_0} = 0 \right)$, thus there exists $\varepsilon(v_0) > 0$ small enough so that $Q \left(\inf_{k \geq 0} W_{k,v_0} \geq 2\varepsilon \right) - Q \left(\inf_{k \geq 0} W_{k,v_0} \leq \varepsilon \right) \geq \frac{1}{2}$. Setting $\varepsilon := \min_{v_0 \in V_0} \varepsilon(v_0)$, we get $\eta > 0$ from (8.5), as desired. $\square$

8.2 A control on the moments

We now prove a generalization of the key Proposition in [JL25], that is, Proposition 7.23. Recall the assumption of bounded weights (BW):

$$\exists C_\mu > 0 \quad \forall x, y \in V \quad C_\mu^{-1} \leq \mu_{x,y} \leq C_\mu.$$

Let us redefine the overshoot stopping time of a threshold, just to stress that it depends on the starting point:

Definition 8.8. Given a starting point $x \in V$ and a threshold $A \geq 1$, the *overshoot stopping time* of A is defined as :

$$\tau_A^x := \min \left\{ n \geq 1 \mid W_{n,x}^\beta \geq A \right\}$$

Then, the Proposition can be extended under (BW):

Proposition 8.9. Assume (BW). Then for any $p \geq 1$, there exists a constant $C_p = C_p(\beta, \omega) > 1$ such that for any $x \in V$ and any $A \geq 1$:

$$Q \left[\left(W_{\tau_A^x}^\beta \right)^p \mid \tau_A^x < +\infty \right] \leq C_p A^p$$

Remark 8.10. Notice that in our generalization, C_p is independent of the starting point $x \in V$.

The proof of Proposition 8.9 is given in Appendix B, using the following notation: Given a random walk S on a weighted graph (G, μ) , recall that the transition matrix is given by $P = (p_{xy})_{x,y \in V} = \left(\frac{\mu_{xy}}{\mu_x} \right)_{x,y \in V}$. For a function $f : V \rightarrow \mathbb{R}$, let us define :

$$\forall x \in V \quad Pf(x) := \sum_{y \sim x} p_{xy} f(y) . \quad (8.6)$$

Then Pf is well defined under (CLF), since the sum is finite in that case. The proof of (8.9) also depends on the following lemma, that holds true in all generality:

Lemma 8.11. *[Lemma 3.1 in [JL25]] Let $(Y_i)_{i \geq 1}$ be a sequence of i.i.d non-negative random variables on a probability space $(X, \mathcal{X}, \mathbb{P})$, with moments of all orders and expectation $\mathbb{E}[Y_1] = 1$. Then for any $q \geq 1$, there exists $C_q \geq 1$ such that for any sequence $(\alpha_i)_{i \geq 1} \in \mathbb{R}_+^{\mathbb{N}}$ satisfying $\sum_{i \geq 1} \alpha_i \leq 1$,*

$$\mathbb{P} \left(\sum_{i \geq 1} \alpha_i Y_i \geq 2 \right) \leq C_q |\alpha|_{\infty}^q .$$

Now, recall that in [JL25], this proposition led to two interesting corollaries. The first was a control of the probability that τ_A is finite and was used to prove Theorem 7.26. The second showed that $p^*(\beta) > 1$ in the weak disorder region, concluding on the chain of equivalences mentioned at the beginning of Section 8.1. Although Proposition 8.9 holds under the mild assumption (BW), those corollaries also depend on Theorem 7.22, which we have only been able to extend as Theorem 8.7 under the quasi-transitivity hypothesis.

Remark 8.12. A quasi-transitive weighted graph satisfies (BW). Indeed, there are a finite number of possible values for the weights. For any $x \in V$, there exists $v_0 \in V_0$ such that $\mu_{x,x} = \mu_{v_0,v_0}$, and for any edge $e = (x, y) \in E$, there exists $v_0 \in V$ and $y' \in N(v_0)$ such that $\mu_{x,y} = \mu_{v_0,y'}$.

Before stating these corollaries for quasi-transitive graphs, we need the following proposition:

Proposition 8.13. *Let (G, μ) be a weighted graph, and $x \in V$. Define*

$$p_x^*(\beta) := \inf \left\{ p \geq 0 \mid \sup_{n \geq 0} Q[(W_{n,x}^\beta)^p] = +\infty \right\} .$$

Then, $p_x^(\beta) =: p^*(\beta)$ does not depend on the starting point $x \in V$.*

Proof. Let $x, y \in V$, and $d := d(x, y)$. Take $p < p^*(\beta)$ so that $(W_{n,x}^\beta)_{n \geq 0}$ is bounded in L^p . Using the recursion formula of Lemma 6.11 for any $n \geq d$, we obtain:

$$W_{n,x}^\beta \geq \widehat{W}_{d,x}^\beta(y) \theta_d^* W_{n-d,y}^\beta .$$

Hence, $Q[(W_{n,x}^\beta)^p] \geq Q[(\widehat{W}_{d,x}^\beta(y))^p] Q[(W_{n-d,y}^\beta)^p]$, so $(W_{n,y}^\beta)_{n \geq 0}$ is also bounded in L^p . Therefore, $p_x^*(\beta) \leq p_y^*(\beta)$. Exchanging the roles of x and y , we also have $p_y^*(\beta) \leq p_x^*(\beta)$. $\square$

We can now state the corollaries of Proposition 8.9.

Corollary 8.14. *Let (G, μ) be quasi-transitive. There exists $c = c_{\beta,Q} > 0$ such that for any starting point $x \in V$ and threshold $A \geq 1$:*

$$Q(\tau_A^x < +\infty) = Q(W_{\infty,x}^{\beta,*} \geq A) \geq c A^{-p^*(\beta)} \quad (8.7)$$

Proof. Since (G, μ) is quasi-transitive, consider $V_0 \subseteq V$ finite satisfying 8.5. From quasi-transitivity and the fact that the environment is i.i.d, given any $x \in V$, there exists $v_0 \in V_0$ such that $\tau_A^x \stackrel{d}{\sim} \tau_A^{v_0}$. Therefore, it is

sufficient to prove (8.7) for any $v_0 \in V_0$.

Fix $(n, v_0) \in \mathbb{N} \times V_0$, and let any $p \geq 1$. In order to make use of Proposition 8.9, let us evaluate :

$$\begin{aligned} Q \left[\left(W_{n,v_0}^\beta \right)^p \right] &= Q \left[\left(W_{n,v_0}^\beta \right)^p \mathbf{1}_{\tau_A^{v_0} > n} \right] + \sum_{k=1}^n Q \left[\left(W_{n,v_0}^\beta \right)^p \mathbf{1}_{\tau_A^{v_0} = k} \right] \\ &\leq A^p + \sum_{k=1}^n Q \left[\left(W_{k,v_0}^\beta \right)^p \mathbf{1}_{\tau_A^{v_0} = k} Q \left[\left(\frac{W_{n,v_0}^\beta}{W_{k,v_0}^\beta} \right)^p \mid \mathcal{F}_k \right] \right]. \end{aligned}$$

Then, recalling (8.1), we get, from Jensen's inequality :

$$Q \left[\left(\frac{W_{n,v_0}^\beta}{W_{k,v_0}^\beta} \right)^p \mid \mathcal{F}_k \right] \leq Q \left[\sum_{y \in V} \mu_{k,v_0}^\beta(y) \left(\theta_k^* W_{n-k,y}^\beta \right)^p \mid \mathcal{F}_k \right] = \sum_{y \in V} \mu_{k,v_0}^\beta(y) Q \left[\left(W_{n-k,y}^\beta \right)^p \right],$$

and since $\left(\left(W_{n,\cdot}^\beta \right)^p \right)_{n \geq 0}$ is a submartingale, $Q \left[\left(W_{n-k,y}^\beta \right)^p \right] \leq Q \left[\left(W_{n,y}^\beta \right)^p \right]$. Now, from quasi-transitivity and the fact that the environment is *i.i.d*, for any $y \in V$ there exists $v \in V_0$ such that $Q \left[\left(W_{n,y}^\beta \right)^p \right] = Q \left[\left(W_{n,v}^\beta \right)^p \right]$. Therefore, setting $S := \sum_{v \in V_0} Q \left[\left(W_{n,v}^\beta \right)^p \right]$ yields :

$$\begin{aligned} Q \left[\left(W_{n,v_0}^\beta \right)^p \right] &\leq A^p + S \sum_{k=1}^n Q \left[\left(W_{n,v_0}^\beta \right)^p \mathbf{1}_{\tau_A^{v_0} = k} \underbrace{\sum_{y \in V} \mu_{k,v_0}^\beta(y)}_{=1} \right] \\ &\leq A^p + S Q \left[\left(W_{n,v_0}^\beta \right)^p \mathbf{1}_{\tau_A^{v_0} < +\infty} \right] \\ &\leq A^p + S C_p A^p Q \left(\tau_A^{v_0} < +\infty \right) \quad (\text{from 8.9}). \end{aligned}$$

Finally, summing this inequality over V_0 yields :

$$S \leq \#V_0 A^p + S C_p A^p \sum_{v_0 \in V_0} Q \left(\tau_A^{v_0} < +\infty \right). \quad (8.8)$$

Taking $p > p^*(\beta) (\geq 1)$, we have $S \xrightarrow{n \rightarrow +\infty} +\infty$, so that (8.8) implies:

$$\sum_{v_0 \in V_0} Q \left(\tau_A^{v_0} < +\infty \right) \geq \frac{1}{C_p} A^{-p}.$$

Then, letting $p \downarrow p^*(\beta)$ yields the result, since $p \mapsto C_p$ is increasing and continuous. Indeed, looking at the proof of Proposition 8.9, there exists $C = C_{\beta,Q} > 0$ such that $C_p = 2^p + C \exp \left(\frac{\lambda(2p\beta)}{2} - p\lambda(\beta) \right)$.

Then, from the definition of $\lambda(\beta) = \log Q \left[e^{\beta \omega(\cdot, \cdot)} \right]$, it can easily be seen that $\beta \mapsto \lambda(\beta)$ is $\mathcal{C}^\infty$ and strictly convex (in particular, $\beta \mapsto \lambda'(\beta)$ is increasing). Consequently, $C : p \mapsto C_p$ is also differentiable and $C'(p) = p2^{p-1} + Cp(\lambda'(2p\beta) - \lambda'(\beta)) \exp \left(\frac{\lambda(2p\beta)}{2} - p\lambda(\beta) \right)$ is positive for any $p \geq 1$. $\square$

Corollary 8.15. *Let (G, μ) be quasi-transitive. Then $p^*(\beta) > 1$ whenever weak disorder holds at β .*

Proof. The proof is identical to ([JL25], Corollary 2.8 – i)). Let any $x \in V$. Since $W_{0,x}^\beta = 1$ a.s, notice that :

$$\begin{aligned} Q \left[\sup_{n \geq 0} W_{n,x}^\beta \right] &= \int_0^\infty Q \left(\sup_{n \geq 0} W_{n,x}^\beta \geq A \right) dA \\ &= 1 + \int_1^\infty Q \left(\sup_{n \geq 0} W_{n,x}^\beta \geq A \right) dA \\ &= 1 + \int_1^\infty Q \left(\tau_A^x < +\infty \right) dA. \end{aligned}$$

Since weak disorder holds at β , the left-hand side is finite from Proposition 8.7. Considering (8.7) implies the necessity of having $p^*(\beta) > 1$. $\square$

8.3 Estimating the tails

In this section, we investigate a generalization of the key theorem in [JL25], that is, Theorem 7.26. The theorem itself relies on three lemmas, two of which can be extended for quasi-transitive graphs, and the last one that remains valid for general weighted graphs. After proving the theorem for quasi-transitive graphs, we discuss an extension of its many corollaries.

Recall from Proposition 8.13 that $p^*(\beta)$ is independent of the starting point. Our objective is to prove the following theorem:

Theorem 8.16. *Let (G, μ) be a quasi-transitive weighted graph. Let $x \in V$ be a starting point. Let $\beta \geq 0$ such that weak disorder holds. Then*

$$Q(W_{\infty,x}^\beta > u) \asymp Q(W_{\infty,x}^{\beta,*} > u) \asymp Q(\widehat{W}_{\infty,x}^{\beta,*} > u) \asymp u^{-p^*(\beta)} .$$

Where, similarly to the case of the SRW in $\mathbb{Z}^d$,

$$\widehat{W}_{n,x}^\beta(y) := \sup_{n \geq 0} W_{n,x}^\beta, \quad \widehat{W}_{\infty,x}^{\beta,*} := \sup_{(n,y) \in \mathbb{N} \times V} \widehat{W}_{n,x}^\beta(y)$$

with $\widehat{W}_{n,x}^\beta(y) := P_x[e^{\beta H_n(\omega, S) - n\lambda(\beta)} \mathbf{1}_{S_n=y}]$. Proving this theorem amounts to proving six inequalities. First, observe the two immediate inequalities are given by

$$W_{\infty,x}^\beta \leq W_{\infty,x}^{\beta,*} \quad \widehat{W}_{\infty,x}^{\beta,*} \leq W_{\infty,x}^{\beta,*} .$$

Then, Corollary 8.14 yields a third inequality. Thus, three inequalities remain to be shown. The first one is given by the following lemma, which holds for general weighted graphs:

Lemma 8.17. *Let (G, μ) be a weighted graph and $x \in V$. Assume that weak disorder holds at $\beta \geq 0$. Then, for any $u > 1$,*

$$Q(\widehat{W}_{\infty,x}^{\beta,*} > u) \leq u^{-p^*(\beta)} .$$

A second one will follow from the next lemma, which does not exactly match an inequality needed for proving Theorem 8.16, but will suffice.

Lemma 8.18. *Let (G, μ) be a quasi-transitive graph, and $x \in V$. There exists a constant $c > 0$ such that, for any $u > 1$,*

$$Q(W_{\infty,x}^\beta > u) \geq cQ(W_{\infty,x}^{\beta,*} > 4u) .$$

Finally, the most difficult inequality to obtain is the one comparing the tails of $W_{\infty,x}^{\beta,*}$ and $\widehat{W}_{\infty,x}^{\beta,*}$. This is accomplished through the following lemma:

Lemma 8.19. *There exists $C > 1$, $\delta > 0$ and a closed set $\mathcal{U} \subseteq [1, +\infty)$ satisfying the "no large gaps" condition:*

$$\forall v \geq 1 [v, Cv] \cap \mathcal{U} \neq \emptyset \tag{8.9}$$

(log $\mathcal{U}$ has no gaps larger than C) such that the following holds:

$$\forall u \in \mathcal{U} \quad Q \left[\max_{y \in V} \mu_{\tau_u^x, x}(y) \geq \delta \mid \tau_u^x < +\infty \right] \geq \delta .$$

Before going through the proofs of these lemmas, let us show how they lead to Theorem 8.16.

Proof of Theorem 8.16. First, let us show that the tails of $W_{\infty,x}^{\beta,*}$ and $\widehat{W}_{\infty,x}^{\beta,*}$ are both comparable to $u \mapsto u^{-p^*(\beta)}$. Lemma 8.17 and Corollary 8.14 already yield two inequalities comparing the two tails, therefore it suffices to show the existence of $\kappa > 0$ such that for all $u \geq 1$:

$$Q(W_{\infty,x}^{\beta,*} > u) \leq \frac{1}{\kappa} u^{-p^*(\beta)} \quad \text{and} \quad Q(\widehat{W}_{\infty,x}^{\beta,*} > u) \geq \kappa u^{-p^*(\beta)} . \tag{8.10}$$

Let $C > 1$, $\delta > 0$ and $\mathcal{U}$ as in Lemma 8.19. Define :

$$u' := \max \{v \in \mathcal{U} \mid v \leq u\} ,$$

$$u'' := \min \{v \in \mathcal{U} \mid \delta v \geq u\} .$$

In particular, from the no large gaps condition (8.9), $u' \geq u/C$ and $u'' \leq Cu/\delta$. Using Lemma 8.19 for any $v \in \mathcal{U}$ and the fact that $\widehat{W}_{\tau_v^x, x}^\beta(y) = W_{\tau_v^x, x}^\beta(y) \mu_{\tau_v^x, x}^\beta(y) \geq v \mu_{\tau_v^x, x}^\beta(y)$, we obtain:

$$Q(\widehat{W}_{\infty, x}^{\beta, *} \geq \delta v) \geq Q\left(\sup_{y \in V} \widehat{W}_{\tau_v^x, x}^\beta(y) \geq \delta v\right) \geq \delta Q(\tau_v^x < +\infty) . \quad (8.11)$$

Therefore,

$$\begin{aligned} Q(\widehat{W}_{\infty, x}^{\beta, *} \geq u) &\geq Q(\widehat{W}_{\infty, x}^{\beta, *} \geq \delta u'') \\ &\geq \delta Q(\tau_{u''}^x < +\infty) \\ &\geq \delta c(u'')^{-p^*(\beta)} && \text{(from Corollary 8.14)} \\ &\geq cC^{-p^*(\beta)} \delta^{1+p^*(\beta)} u^{-p^*(\beta)} && \text{(from the no large gaps condition)} . \end{aligned}$$

On the other hand,

$$\begin{aligned} Q(W_{\infty, x}^{\beta, *} \geq u) &\leq Q(\tau_{u'}^x < +\infty) \\ &\leq \delta^{-1} Q(\widehat{W}_{\infty, x}^{\beta, *} \geq \delta u') && \text{(from (8.11))} \\ &\leq \delta^{-1-p^*(\beta)} (u')^{-p^*(\beta)} && \text{(from Lemma 8.17)} \\ &\leq C^{p^*(\beta)} \delta^{-1-p^*(\beta)} u^{-p^*(\beta)} && \text{(from the no large gaps condition)} . \end{aligned}$$

These two series of inequalities prove (8.10). All that remains to be done is to show that the tail of $W_{\infty, x}^\beta$ is comparable to $u \mapsto u^{-p^*(\beta)}$. Using the last series of inequalities and the fact that $W_{\infty, x}^\beta \leq W_{\infty, x}^{\beta, *}$ yields one equality. The second one is obtained combining Lemma 8.18 and Corollary 8.14. $\square$

Now, let us focus on the proofs of the different lemmas. Lemma 8.17 directly holds for any (CLF) weighted graph carrying out the same proof as in [JL25]. Hence, we refer to [JL25, Lemma 4.1]. The proof of Lemma 8.18 can be found in Appendix B.

Lemma 8.19 follows directly from [JL25, Lemma 4.7 and Proposition 4.5], whose proofs are directly adaptable to the context of general graphs, except for the end of Proposition 4.5 which constitutes the following lemma:

Lemma 8.20. *Define, for $\eta > 0$,*

$$\Gamma(\eta) := \sup_{\alpha \in \mathcal{M}_1(\mathbb{Z}^d), \|\alpha\|_\infty \leq \eta} Q\left(\sup_{n \geq 0} \sum_{y \in V} \alpha(y) W_{n, y}^\beta \geq 2\right) .$$

If (G, μ) is quasi-transitive, then $\Gamma(\eta) \xrightarrow{\eta \rightarrow +\infty} 0$.

We prove this lemma at the end of Appendix B. To finish this part, remember that Theorem 7.26 yielded a lot of corollaries for the SRW in $\mathbb{Z}^d$. While the proof of Corollary 7.30 is immediately adaptable to our context, we have not yet been able to tackle Corollary 7.28 and [JL25, Corollary 2.11], which constitutes Corollary 7.27 and Corollary 7.29.

9 Conclusion

9.1 Scientific conclusion

In conclusion, the directed polymer is a rich model, starting from a simple definition and exhibiting complex properties and applications to physics and other areas of mathematics. The field of its study has attracted many researchers at the frontier between probability theory, real analysis, partial and stochastic differential equations theory, and mathematical physics. Very recently, Stefan Junk and Hubert Lacoin have been able to make some remarkable breakthroughs regarding fundamental questions about the behavior of the polymer at criticality. Their works have used a very impressive range of tools and techniques from probability and analysis, as well as links with other related models such as the pinning model.

For a long time, mathematicians have almost exclusively studied the case where the polymer is initially a simple random walk in $\mathbb{Z}^d$, then perturbed by a random environment. Of course, from a physical perspective, the simple random walk in $\mathbb{Z}^d$ might not be suited to model a polymer chain in a given medium: such a polymer could evolve in a fractal network, such as human lungs, and could have a natural preference for particular directions or regions of its space, before considering any obstacle or reward. Hence, some early works have given some results on the polymer model for specific graphs, for instance trees, and often included a physicist among the authors.

It is only recently that two articles have brought to light attempts to study the model for other random walks and graphs. One of them [KKN20] counted one of my supervisors, Naotaka Kajino, as a co-author. Therefore, our objective has been to continue extending the model to general weighted graphs, in particular the recent works of Stefan Junk and Lacoin. As mentioned before, this has turned out to be a quite difficult challenge. Most of Junk and Lacoin's proofs relied on general techniques from probability theory and analysis on positive martingales. The feature of the model that was primarily used was the recursion formula of Lemma 6.11, which could be dealt with using the symmetry of the problem: symmetry of the graph (each vertex "plays the same role"), the weights (which are identical on all edges) and the environment that is i.i.d..

Hence, during this internship, we have identified the arguments for which the choice of $\mathbb{Z}^d$ as the weighted graph was important. Some lemmas could be easily extended to general weighted graphs. Notably, the key proposition, that is Proposition 7.23, has been generalized to any weighted graph satisfying the very mild condition (BW). Other results of [JL25] were much more dependent on the symmetric structure of $\mathbb{Z}^d$. In the allocated time for the internship, we have only succeeded to extend these results to quasi-transitive graphs, an interesting class of graphs that contains periodic tilings and Cayley graphs, but excludes typical fractals that were our initial subject of interest.

The general weighted graphs approach of the model raises many new questions. In $\mathbb{Z}^d$, the influence of the structure of the lattice was implicit and the study was driven towards understanding nonnegative martingales with a certain multiplicative structure given by the recursion formula and the underlying space. In the general case, many properties of the underlying weighted graph should need to be scrutinized in order to understand their influence on the polymer: their connectivity, asymmetry, and the existence of concentrated or preferred regions in terms of cycles or conductance. Studying random walks on infinite graphs, especially fractal ones, can also require classical tools from analysis on fractal graphs. For instance, considering random walks that satisfy the heat kernel estimates we presented (see (UHK) and (LHK)) is usual, and the study of such random walks has been conducted mostly in the partial differential equations community as well as in Riemannian geometry.

Hence, studying the polymer model in the broader context of general weighted graphs further highlights its richness and depth, connecting the model with even more mathematical disciplines, namely graph theory, analysis on fractals and Riemannian geometry.

9.2 Personal conclusion

This internship has been a particularly rewarding experience, both mathematically and personally. I greatly enjoyed working on a research project in probability and discovering the work of mathematical research.

I had the opportunity to see how the results and techniques encountered during my probability courses

this year can actually be used to approach current research problems. This gave me a much more concrete understanding of the role of these theorems. At the same time, the internship allowed me to deepen my knowledge of probability theory, while also introducing me to other areas of mathematics with which I was less familiar. In particular, conducting a state-of-the-art study covering the history and development of the model was extremely instructive.

I also greatly appreciated discovering the environment of a mathematical research laboratory. Through discussions with my supervisors, office colleagues, and other researchers, I was able to get a much clearer idea of what the daily life of a mathematician looks like. I also had the opportunity to attend many seminars and conferences. In particular, I participated in a three-day workshop on heat kernel estimates and attended lectures at the RIMS, the Mathematics department of Kyoto University, and on two occasions at the University of Osaka. One of the talks in Osaka was given by Clément Cosco, one of the authors of [CSZ21], who currently works in Paris and with whom I hope to remain in contact. I also had the opportunity to attend a mini-course given by one of my supervisors, Hélène Guérin. These experiences allowed me to discover many new mathematical topics, and the way researchers present their talks in English.

Another important aspect of the internship was the improvement of my communication skills. In particular, I became more comfortable speaking in English and improved my ability to explain complex mathematical ideas clearly. Every week, I presented to my supervisors what I had read and what I had been working on. Since the subject of the internship was particularly close to the research interests of one of my supervisors, he could examine my arguments very carefully, while I also had to explain the papers I was studying to my other supervisor, who was discovering this particular topic at the same time. This was a very useful exercise: having to explain an argument to someone who was not already familiar with it forced me to understand the papers thoroughly rather than relying on implicit steps or arguments that I had not fully understood. At the same time, the more experienced and critical perspective of my supervisor often led her to ask questions that I had not thought about myself, helping me identify gaps in my understanding.

Finally, giving a final presentation at the end of the internship was a particularly valuable experience. Presenting my work during a seminar and entirely in English taught me how to present a mathematical subject in a way that is both comprehensive and understandable.

Overall, this internship has strengthened my interest in probability and in mathematical research in general. Beyond the mathematical knowledge I gained, it gave me a clearer picture of what pursuing a career in research would involve and provided me with valuable experience of working within an international mathematical community. As the problem was really interesting and challenging, I wish I had more time for the internship to deepen my work and explore some ideas. Therefore, I plan to remain in contact with my supervisors and Clément Cosco, and next year I will attend a course on the directed polymer model taught by Quentin Berger (a co-author in [BT10]) at Sorbonne Université's *M2 Probabilités et Modèles Aléatoires*. The course will cover both the topics discussed in this report and the connections between the model and the stochastic heat equation with multiplicative noise.

In conclusion, I am particularly grateful to my supervisors for their attentive guidance and for always being there to support me throughout the internship. I also would like to thank the RIMS administration for their warm welcome and for their great support throughout my stay in Japan, as well as for the excellent working conditions I have been given. Finally, I would like to thank the Mathematics Department at ENS Paris-Saclay for having given me the opportunity to do this internship.

A Appendix : general results of probability theory

In this appendix, we give some general results of probability theory that have been used to prove Theorem 6.9.

Proposition A.1 (La Vallée-Poussin criterion). *Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be a monotone increasing function such that $x^{-1}\phi(x) \rightarrow +\infty$ as $x \rightarrow +\infty$. Then, for any $C \geq 0$, the set $\{X \in L^1(\Omega, \mathcal{F}, Q) \mid Q[|X|] \leq C\}$ is uniformly integrable.*

Proof. Indeed, notice that

$$\begin{aligned} Q[|X| \mathbf{1}_{|X|>a}] &= Q\left[\frac{|X|}{\phi(|X|)} \phi(|X|) \mathbf{1}_{|X|>a}\right] \\ &\leq \left(\sup_{x>a} \frac{x}{\phi(x)}\right) Q[\phi(|X|)] \\ &\leq \left(\sup_{x>a} \frac{x}{\phi(x)}\right) C, \end{aligned}$$

and the right-hand side goes to 0 as $a \rightarrow +\infty$. $\square$

Corollary A.2. *The sequence $((W_{n,x}^\beta)^\theta)_{n \geq 0}$ is uniformly integrable for any $\theta \in (0, 1)$.*

Proof. Using the criterion with $\phi(x) := x^{1/\theta}$ yields the result, since $Q[W_{n,x}^\beta] = 1$. $\square$

Proposition A.3 (Lebesgue-Vitali). *Let $(X_n)_{n \geq 0}$ be a sequence of L^1 random variables such that $X_n \xrightarrow{\mathbb{P}} X_\infty$. If $(X_n)_{n \geq 0}$ is uniformly integrable, then it converges to X_∞ in L^1 .*

Remark A.4. The converse also holds, though we do not need it here.

Proof. From the triangular inequality, the sequence $(X_n - X_m)_{n,m \geq 0}$ is itself uniformly integrable. Given $\varepsilon > 0$, consider $a > 0$ large enough so that

$$Q[|X_n - X_m| \mathbf{1}_{|X_n - X_m|>a}] \leq \varepsilon.$$

Then, for any $n, m \geq 0$, we have

$$\begin{aligned} Q[|X_n - X_m|] &= Q[|X_n - X_m| \mathbf{1}_{|X_n - X_m| \leq \varepsilon}] + Q[|X_n - X_m| \mathbf{1}_{\varepsilon < |X_n - X_m| \leq a}] + Q[|X_n - X_m| \mathbf{1}_{|X_n - X_m| > a}] \\ &\leq 2\varepsilon + aQ(|X_n - X_m| > \varepsilon). \end{aligned}$$

Since $(X_n)_{n \geq 0}$ converges in probability to X_∞ , the triangular inequality yields

$$Q(|X_n - X_m| > \varepsilon) \leq Q(|X_n - X_\infty| > \varepsilon/2) + Q(|X_m - X_\infty| > \varepsilon/2) \xrightarrow{n,m \rightarrow +\infty} 0,$$

therefore $\limsup_{n,m \rightarrow +\infty} Q[|X_n - X_m|] \leq 2\varepsilon$. Since $\varepsilon > 0$ is arbitrary, it follows that $(X_n)_{n \geq 0}$ is a Cauchy sequence for the L^1 -norm. $\square$

Proposition A.5 (FKG inequality). *Let $(X_1, \dots, X_n)$ be independent real-valued random variables. Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be non-decreasing functions, where $\mathbb{R}^n$ is endowed with the coordinates order:*

$$\forall x, y \in \mathbb{R}^n \quad x \leq y \iff \forall i \in \llbracket 1, n \rrbracket, \quad x_i \leq y_i.$$

Then

$$Q[f(X_1, \dots, X_n)g(X_1, \dots, X_n)] \geq Q[f(X_1, \dots, X_n)]Q[g(X_1, \dots, X_n)].$$

Proof. Let us proceed by induction on $n \geq 1$. For any $n = 1$, let $X := X_1$ and X' be an independent copy of X . On the one hand :

$$Q[f(X)g(X)] - Q[f(X)]Q[g(X)] = Q[f(X)(g(X) - g(X'))],$$

and on the other hand,

$$Q[f(X)g(X)] - Q[f(X)]Q[g(X)] = Q[f(X')g(X')] - Q[f(X')]Q[g(X')] = Q[f(X')(g(X') - g(X))],$$

hence $Q[f(X)g(X)] - Q[f(X)]Q[g(X)] = \frac{1}{2}Q[(f(X) - f(X'))(g(X) - g(X'))]$. However, since f and g are both non-decreasing,

$$\forall x, x' \in \mathbb{R} \quad (f(x) - f(x'))(g(x) - g(x')) \geq 0,$$

thus $Q[f(X)g(X)] \geq Q[f(X)]Q[g(X)]$.

Now, assume that the proposition holds for any $n-1$ tuple of random variables. Let $X := (X_1, \dots, X_n)$ be independent random variables. Set :

$$\begin{aligned} F(x_1, \dots, x_{n-1}) &:= Q[f(x_1, \dots, x_{n-1}, X_n)] \\ G(x_1, \dots, x_{n-1}) &:= Q[g(x_1, \dots, x_{n-1}, X_n)] . \end{aligned}$$

Since f and g are non-increasing, F and G are non-increasing as well. For fixed values of $X_1, \dots, X_{n-1}$, the base case yields

$$Q[f(X)g(X) \mid X_1, \dots, X_{n-1}] \geq F(X_1, \dots, X_{n-1})G(X_1, \dots, X_{n-1}) .$$

Then, since F and G are non-increasing and $X_1, \dots, X_{n-1}$ are independent, the induction hypothesis gives

$$Q[F(X_1, \dots, X_{n-1})G(X_1, \dots, X_{n-1})] \geq Q[F(X_1, \dots, X_{n-1})]Q[G(X_1, \dots, X_{n-1})] = Q[f(X)]Q[g(X)] ,$$

hence $Q[f(X)g(X)] \geq Q[f(X)]Q[g(X)]$. $\square$

B Appendix : some proofs of our personal developments

In this second appendix, we give the proofs of Proposition 8.9, Lemma 8.18, and Lemma 8.20

Proof (Proposition 8.9). If $\text{ess sup } \omega(1, 0) < +\infty$, then the proposition holds with $C_p := L^p$. Therefore, let us assume that $\omega(1, 0)$ is unbounded from above. Set $\tau := \tau_A^x$ for simplicity. First, let us distinguish the "small overshoot" case from the large one :

$$\begin{aligned} Q \left[(W_{\tau,x}^\beta)^p \mathbf{1}_{\tau < +\infty} \right] &= Q \left[(W_{\tau,x}^\beta)^p \mathbf{1}_{\tau < +\infty} \left(\mathbf{1}_{A \leq W_{\tau,x}^\beta \leq 2A} + \mathbf{1}_{W_{\tau,x}^\beta \geq 2A} \right) \right] \\ &\leq (2A)^p Q(\tau < +\infty) + Q \left[(W_{\tau,x}^\beta)^p \mathbf{1}_{\tau < +\infty, W_{\tau,x}^\beta \geq 2A} \right] . \end{aligned}$$

Next, decomposing over the possible values of τ , we get :

$$\begin{aligned} Q \left[(W_{\tau,x}^\beta)^p \mathbf{1}_{\tau < +\infty, W_{\tau,x}^\beta \geq 2A} \right] &= \sum_{n \geq 1} Q \left[(W_{n,x}^\beta)^p \mathbf{1}_{\tau=n, W_{n,x}^\beta \geq 2A} \right] \\ &= \sum_{n \geq 1} Q \left[(W_{n,x}^\beta)^p \mathbf{1}_{\tau \geq n, W_{n,x}^\beta \geq 2A} \right] \quad (\text{on } \{W_{n,x}^\beta \geq 2A\}, \tau = n \text{ iff } \tau \geq n .) \\ &= \sum_{n \geq 1} Q \left[Q \left[(W_{n,x}^\beta)^p \mathbf{1}_{W_{n,x}^\beta \geq 2A} \mid \mathcal{F}_{n-1} \right] \mathbf{1}_{\tau \geq n} \right] . \end{aligned}$$

Then, from Cauchy-Schwarz's inequality :

$$Q \left[(W_{n,x}^\beta)^p \mathbf{1}_{W_{n,x}^\beta \geq 2A} \mid \mathcal{F}_{n-1} \right] \leq \sqrt{Q \left[(W_{n,x}^\beta)^{2p} \mid \mathcal{F}_{n-1} \right] Q \left(W_{n,x}^\beta \geq 2A \mid \mathcal{F}_{n-1} \right)} .$$

Now, recalling the definitions of Pf (8.6) and $\mu_{n,x}^\beta(y)$ (see Lemma 8.6), the following holds :

$$W_{n,x}^\beta = \sum_{y \in V} P\widehat{W}_{n-1,x}(y)\xi_y = W_{n-1,x} \sum_{y \in V} P\mu_{n-1,x}^\beta(y)\xi_y ,$$

with $\xi_y := e^{\beta\omega(n,y) - n\lambda(\beta)}$. Notice that the sum is actually taken over $B(x, n)$, which is finite under (CLF). Hence, the right-hand side is well defined. The random variables $(\xi_y)_{y \in V}$ are i.i.d., independent of $\mathcal{F}_{n-1}$, have

moments of all orders, and expectation $Q[\xi_y] = 1$. The coefficients $\left(P\mu_{n-1,x}^\beta(y)\right)_{y \in V}$ are $\mathcal{F}_{n-1}$ -measurable and sum to 1. Therefore, on the event $\{\tau \geq n\}$,

$$\begin{aligned} Q \left[(W_{n,x}^\beta)^{2p} \mid \mathcal{F}_{n-1} \right] &= (W_{n-1,x}^\beta)^{2p} Q \left[\left(\sum_{y \in V} D\mu_{n-1,x}^\beta(y) \xi_y \right)^{2p} \mid \mathcal{F}_{n-1} \right] \\ &\leq A^{2p} Q \left[\sum_{y \in V} D\mu_{n-1,x}^\beta(y) \xi_y^{2p} \mid \mathcal{F}_{n-1} \right] \quad (\text{Jensen's inequality}) \\ &\leq A^{2p} \sum_{y \in V} D\mu_{n-1,x}^\beta(y) Q[\xi_y^{2p}] \\ &= A^{2p} e^{\lambda(2p\beta) - 2p\lambda(\beta)} . \end{aligned}$$

Consequently, all that remains to show is that there exists $C > 0$ such that

$$\sum_{n \geq 1} Q \left[\sqrt{Q(W_{n,x}^\beta \geq 2A \mid \mathcal{F}_{n-1})} \mathbf{1}_{\tau \geq n} \right] \leq C Q(\tau < +\infty) . \quad (\text{B.1})$$

Using Lemma 8.11 with $Y_y := \xi_y$ and $\alpha_y := \frac{1}{A} P\widehat{W}_{n-1,x}^\beta(y)$, we obtain, for any $q \in \mathbb{N}^*$ (on the event $\{\tau \geq n\}$ that ensures $W_{n-1,x}^\beta \leq A$ so that $\sum_{y \in V} \alpha_y \leq 1$):

$$\begin{aligned} \exists C_q > 0 \quad Q(W_{n,x}^\beta \geq 2A) &\leq C_q \left(\max_{y \in V} A^{-1} D\widehat{W}_{n-1,x}^\beta(y) \right)^q \\ &\leq C_q \left(\max_{y \in V} A^{-1} \widehat{W}_{n-1,x}^\beta(y) \right)^q . \end{aligned}$$

Now, set $\kappa := Q \left(e^{\beta\omega(\cdot, \cdot) - \lambda(\beta)} \geq \frac{2}{\tilde{p}} \right) \wedge \frac{1}{2}$ with :

$$\tilde{p} := \inf_{x \in V} \max_{y \sim x} p_{xy} , \quad (\text{B.2})$$

which is positive under (BW). Note that $\kappa > 0$ from our assumption that $\omega(\cdot, \cdot)$ is unbounded from above. This choice for κ allows the inequality :

$$Q \left(W_{n+1+k,x}^\beta \geq 2^k \max_{y \in V} \widehat{W}_{n-1,x}^\beta(y) \mid \mathcal{F}_{n-1} \right) \quad (\text{B.3})$$

to hold. Indeed, set $y_0 := \arg \max \widehat{W}_{n-1,x}^\beta(y)$ and construct the path $\{y_1, \dots, y_k\}$ by induction, choosing $y_{i+1} \in N(y_i)$ such that $p(y_i, y_{i+1}) \geq \tilde{p}$. Such a choice for y_{i+1} is made possible by (B.2). On the event $\mathcal{E} := \left\{ \forall i \in \llbracket 1, k \rrbracket \quad e^{\beta\omega(n-1+i, y_i) - \lambda(\beta)} \geq \frac{2}{\tilde{p}} \right\}$, the event on the left-hand side of (B.3) is realized, since :

$$\begin{aligned} W_{n+1+k,x}^\beta &= P_x \left[e^{\beta H_{n-1+k,x}(\omega, S) - (n-1+k)\lambda(\beta)} \right] \\ &\geq P_x \left[e^{\beta H_{n-1+k,x}(\omega, S) - (n-1+k)\lambda(\beta)} \mathbf{1}_{S_{n-1}=y_0, S_n=y_1, \dots, S_{n-1+k}=y_k} \right] \\ &= P_x \left[e^{\beta H_{n-1,x}(\omega, S) - n\lambda(\beta)} \mathbf{1}_{S_{n-1}=y_0} \right] P_{y_0} \left[e^{\beta \theta_{n-1}^* H_{k,y_0}(\omega, S) - k\lambda(\beta)} \mathbf{1}_{S_n=y_1, \dots, S_{n-1+k}=y_k} \right] \\ &\geq \widehat{W}_{n-1,x}^\beta(y_0) \tilde{p}^k \prod_{i=1}^k e^{\beta\omega(n-1+i, y_i) - \lambda(\beta)} \\ &\geq \widehat{W}_{n-1,x}^\beta(y_0) \tilde{p}^k \left(\frac{2}{\tilde{p}} \right)^k \\ &= 2^k \widehat{W}_{n-1,x}^\beta(y_0) . \end{aligned}$$

The passage from line 2 to line 3 comes from Markov's property. The probability of $\mathcal{E}$ is exactly κ^k . Now, (B.3) still holds if k is replaced by any $\mathcal{F}_{n-1}$ -measurable random variable. Ingeniously setting

$$\Theta_n := \left\lceil \log_2 \left(\frac{A}{\max_{y \in V} \widehat{W}_{n-1,x}^\beta(y)} \right) \right\rceil \quad (\text{B.4})$$

yields :

$$\kappa^{\Theta_n} \leq Q \left(W_{n+1+\Theta_n,x}^\beta \geq 2^{\Theta_n} \max_{y \in V} \widehat{W}_{n-1,x}^\beta(y) \mid \mathcal{F}_{n-1} \right) \leq Q \left(W_{n+1+\Theta_n,x}^\beta \geq A \mid \mathcal{F}_{n-1} \right) ,$$

so that, on the event $\{\tau \geq n\}$:

$$1 \leq \kappa^{-\Theta_n} Q \left(W_{n+1+\Theta_n,x}^\beta \geq A \mid \mathcal{F}_{n-1} \right) \leq \kappa^{-\Theta_n} Q \left[\mathbb{1}_{[n,n-1+\Theta_n]}(\tau) \mid \mathcal{F}_{n-1} \right] .$$

Hence, setting $q := -4 \log_2 \kappa$ so that $2^{q/2} \kappa = \kappa^{-1}$, (B.1) can be upper-bounded as follows :

$$\begin{aligned} Q \left[\sqrt{Q \left(W_{n,x}^\beta \geq 2A \mid \mathcal{F}_{n-1} \right)} \mathbb{1}_{\tau \geq n} \right] &\leq \sqrt{C_q} Q \left[\left(\max_{y \in V} A^{-1} \widehat{W}_{n-1,x}^\beta(y) \right)^{q/2} \mathbb{1}_{\tau \geq n} \right] \\ &\leq \sqrt{C_q} Q \left[\mathbb{1}_{\tau \geq n} (2^{1-\Theta_n})^{q/2} \times 1 \right] \quad (\text{from (B.4)}) \\ &\leq \sqrt{C_q} Q \left[\mathbb{1}_{\tau \geq n} (2^{1-\Theta_n})^{q/2} \kappa^{-\Theta_n} Q \left[\mathbb{1}_{[n,n-1+\Theta_n]}(\tau) \mid \mathcal{F}_{n-1} \right] \right] \\ &= \sqrt{C_q} 2^{q/2} Q \left[\mathbb{1}_{[n,n-1+\Theta_n]}(\tau) \kappa^{\Theta_n} \right] \\ &\leq \sqrt{C_q} 2^{q/2} Q \left[\mathbb{1}_{n \leq \tau < +\infty} \kappa^{1+\tau-n} \right] , \end{aligned}$$

where the second-to-last equation is given by the fact that Θ_n and $\mathbb{1}_{\tau \geq n}$ are $\mathcal{F}_{n-1}$ -measurable, and the conditional expectation leaves the mean value unchanged.

Since $\kappa \leq \frac{1}{2}$, it holds that $\sum_{n \geq 1} \kappa^{1+\tau-n} \mathbb{1}_{n \leq \tau < +\infty} \leq \mathbb{1}_{\tau < +\infty}$, so that :

$$\sum_{n \geq 1} Q \left[\sqrt{Q \left(W_{n,x}^\beta \geq 2A \mid \mathcal{F}_{n-1} \right)} \mathbb{1}_{\tau \geq n} \right] \leq \sqrt{C_q} 2^{q/2} Q(\tau < +\infty) ,$$

which concludes from (B.1). □

Proof of Lemma 8.18. Denote $V_0 \subseteq V$ the finite set making (G, μ) quasi-transitive. Without loss of generality, let us assume $x \in V_0$. Set $A := 4u$. On the event $\tau_A^x < +\infty$, the recursion formula yields

$$W_{\infty,x}^\beta = W_{\tau_A^x,x}^\beta \sum_{y \in V} \mu_{\tau_A^x,x}^\beta(y) \theta_{\tau_A^x} W_{\infty,y}^\beta \geq A \sum_{y \in V} \mu_{\tau_A^x,x}^\beta(y) \theta_{\tau_A^x} W_{\infty,y}^\beta ,$$

where $\mu_{\tau_A^x,x}^\beta(y)$ is $\mathcal{F}_{\tau_A^x}^x$ -measurable, and $\theta_{\tau_A^x} W_{\infty,y}^\beta$ is independent from $\mathcal{F}_{\tau_A^x}^x$. Hence, for any $u \geq 0$, the following holds:

$$Q(W_{\infty,x}^\beta \geq u) \geq Q(\tau_A^x < +\infty) \inf_{\alpha \in \mathcal{M}_1(\mathbb{Z}^d)} Q \left(\sum_{y \in V} \alpha(y) \theta_{\tau_A^x} W_{\infty,y}^\beta \geq \frac{1}{4} \right) =: Q(\tau_A^x < +\infty) \eta .$$

To conclude, all we have to do is to show that the infimum defining η is positive. Setting

$$\mathcal{W}_\alpha := \sum_{y \in V} \alpha(y) \theta_{\tau_A^x} W_{\infty,y}^\beta ,$$

we will show that the family $(\mathcal{W}_\alpha)_{\alpha \in \mathcal{M}_1(\mathbb{Z}^d)}$ is uniformly integrable. Consider, for any $v \in V_0$, $\varphi_v : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ convex such that $Q[\varphi_v(W_{\infty,v}^\beta)] < +\infty$ and $\frac{\varphi_v(x)}{x} \xrightarrow{x \rightarrow +\infty} +\infty$. Such a function exists since $W_{\infty,v}^\beta$ is L^1 , take

for example $\varphi_v(x) := \sum_{k \geq 0} (x - n_k^v)^+$, where $(n_k^v)_{k \geq 0}$ is a diverging sequence of natural integers chosen to satisfy $Q[(W_{\infty,v}^\beta - n_k^v)^+] \leq k^{-2}$. Then, set $v_0 := \arg \max_{v \in V_0} Q[\varphi_v(W_{\infty,v}^\beta)]$, and $\psi := \min_{v \in V_0} \varphi_v$. Also, denote $\varphi_y := \varphi_v$ for any $y \in V$ and $v \in V_0$ such that $(G, \mu, y) \cong (G, \mu, v)$. Now, ψ is still convex, satisfies $Q[\psi(W_{\infty,y}^\beta)] < +\infty$ for any $y \in V$ (we can assume $y \in V_0$, and use that $\psi \leq \varphi_y$) and $\psi(x)/x \rightarrow +\infty$. Furthermore:

$$\begin{aligned} Q[\psi(W_\alpha)] &\leq \sum_{y \in V} \alpha(y) Q[\psi(W_{\infty,y}^\beta)] && \text{(Jensen's inequality)} \\ &\leq \sum_{y \in V} \alpha(y) Q[\varphi_y(W_{\infty,y}^\beta)] \\ &\leq Q[\varphi_{v_0}(W_{\infty,v_0}^\beta)] < +\infty. \end{aligned}$$

Hence, applying La Vallée-Poussin criterion to ψ (see Proposition A.1 in Appendix A), we obtain the desired uniform integrability. Since $Q[W_\alpha] = 1$, using this uniform integrability gives the existence of $M > 0$ large enough so that

$$\begin{aligned} Q[W_\alpha \mathbf{1}_{1/4 \leq W_\alpha \leq M}] &= 1 - Q[W_\alpha \mathbf{1}_{W_\alpha < 1/4}] - Q[W_\alpha \mathbf{1}_{W_\alpha > M}] \\ &\geq \frac{3}{4} - Q[W_\alpha \mathbf{1}_{W_\alpha > M}] \geq \frac{1}{2}. \end{aligned}$$

Thus,

$$Q[W_\alpha \geq 1/4] \geq \frac{1}{M} Q[W_\alpha \mathbf{1}_{1/4 \leq W_\alpha \leq M}] \geq \frac{1}{2M}.$$

□

Proof of Lemma 8.20. Since $W_{0,x}^\beta = 1$ a.s., $\sup_{n \geq 0} \sum_{y \in V} \alpha(y) W_{n,y}^\beta - 1$ is nonnegative a.s. so that Markov's inequality yields

$$\Gamma(\eta) \leq \sup_{\alpha \in \mathcal{M}_1(\mathbb{Z}^d), |\alpha|_\infty \leq \eta} Q \left[\sup_{n \geq 0} \sum_{y \in V} \alpha(y) W_{n,y}^\beta - 1 \right].$$

Next, we aim to reduce the range of the supremum over $n \geq 0$. Introducing a (fixed for now) integer $M \geq 0$, notice that

- If $0 \leq n \leq M$, $\sum_{y \in V} \alpha(y) W_{n,y}^\beta \leq \sup_{0 \leq j \leq M} \sum_{y \in V} \alpha(y) W_{j,y}^\beta$.
- If $n > M$,

$$\begin{aligned} \sum_{y \in V} \alpha(y) W_{n,y}^\beta &= \sum_{y \in V} \alpha(y) W_{M,y}^\beta + \sum_{y \in V} \alpha(y) (W_{n,y}^\beta - W_{M,y}^\beta) \\ &\leq \sup_{0 \leq j \leq M} \sum_{y \in V} \alpha(y) W_{j,y}^\beta + \sum_{y \in V} \alpha(y) \sup_{i \geq 0} (W_{M+i,y}^\beta - W_{M,y}^\beta) \end{aligned}$$

Therefore, we obtain

$$Q \left[\sup_{n \geq 0} \sum_{y \in V} \alpha(y) W_{n,y}^\beta \right] \leq Q \left[\sup_{0 \leq j \leq M} \sum_{y \in V} \alpha(y) W_{j,y}^\beta \right] + \sum_{y \in V} \alpha(y) Q \left[\sup_{i \geq 0} W_{M+i,y}^\beta - W_{M,y}^\beta \right].$$

On the one hand, using that α is a probability measure and letting $V_0 \subseteq V$ be the set making (G, μ) quasi-transitive, we get

$$\sum_{y \in V} \alpha(y) Q \left[\sup_{i \geq 0} W_{M+i,y}^\beta - W_{M,y}^\beta \right] \leq \max_{v \in V_0} Q \left[\sup_{i \geq 0} W_{M+i,v}^\beta - W_{M,v}^\beta \right].$$

Now, since $(W_{n,v}^\beta)_{n \geq 0}$ converges a.s. for any $v \in V_0$, we have

$$\lim_{M \rightarrow +\infty} \left(\sup_{i \geq 0} W_{M+i,v}^\beta - W_{M,v}^\beta \right) = \limsup W_{M,v}^\beta - W_{\infty,v}^\beta = 0 .$$

Recall that $W_{\infty,v}^\beta \in L^1$ from Proposition 8.7, and $\sup_{i \geq 0} |W_{M+i,v}^\beta - W_{M,v}^\beta| \leq 2W_{\infty,v}^{\beta,*}$. Hence, given $\varepsilon > 0$, the dominated convergence theorem yields, for M large enough,

$$\max_{v \in V_0} Q \left[\sup_{i \geq 0} W_{M+i,v}^\beta - W_{M,v}^\beta \right] \leq \varepsilon/2 .$$

On the other hand,

$$Q \left[\sup_{0 \leq j \leq M} \sum_{y \in V} \alpha(y) W_{j,y}^\beta - 1 \right] \leq Q \left[\sup_{0 \leq j \leq M} \left(\sum_{y \in V} \alpha(y) W_{j,y}^\beta - 1 \right)^2 \right]^{1/2} \quad (\text{Cauchy-Schwarz})$$

$$\leq 2Q \left[\left(\sum_{y \in V} \alpha(y) W_{M,y}^\beta - 1 \right)^2 \right]^{1/2} \quad (\text{Doob's } L^2\text{-inequality})$$

$$= 2 \left(\sum_{y,y' \in V} \alpha(y) \alpha(y') Q \left[(W_{M,y}^\beta - 1)(W_{M,y'}^\beta - 1) \right] \right)^{1/2} .$$

Now, for $y, y' \in V$ such that $d(y, y') > 2M$, $W_{M,y}^\beta$ and $W_{M,y'}^\beta$ are independant, thus

$$Q \left[(W_{M,y}^\beta - 1)(W_{M,y'}^\beta - 1) \right] = Q \left[W_{M,y}^\beta - 1 \right] Q \left[W_{M,y'}^\beta - 1 \right] = 0 .$$

Therefore, we can restrict the summation on $\{(y, y') \in V^2 \mid d(y, y') \leq 2M\}$, and since

$$\begin{aligned} Q \left[(W_{M,y}^\beta - 1)(W_{M,y'}^\beta - 1) \right] &\leq Q[(W_{M,y}^\beta - 1)^2]^{1/2} Q[(W_{M,y'}^\beta - 1)^2]^{1/2} && (\text{Cauchy-Schwarz}) \\ &\leq Q[(W_{M,y}^\beta)^2]^{1/2} Q[(W_{M,y'}^\beta)^2]^{1/2} && (\text{since } Q[W_{M,y}^\beta] = Q[W_{M,y'}^\beta] = 1) \\ &\leq e^{M(\lambda(2\beta) - 2\lambda(\beta))} && (\text{Jensen's inequality and Fubini}) , \end{aligned}$$

we obtain, for a fixed (but arbitrary) $y \in V$:

$$Q \left[\sup_{0 \leq j \leq M} \sum_{y \in V} \alpha(y) W_{j,y}^\beta - 1 \right] \leq 2 \left(\#B(y, 2M) \underbrace{|\alpha|_\infty}_{\leq \eta} e^{M(\lambda(2\beta) - 2\lambda(\beta))} \right)^{1/2} . \quad (\text{B.5})$$

Hence, taking $\eta \leq \frac{\varepsilon^2}{16 \#B(y, 2M)} e^{-M(\lambda(2\beta) - 2\lambda(\beta))}$, so that the left-hand side of (B.5) is upper-bounded by $\varepsilon/2$, we get

$$\Gamma(\eta) \leq \varepsilon ,$$

which concludes the proof. $\square$

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